

Diophantine Confinement in Syracuse Dynamics: Computational Evidence and Extended Analysis

John A. Janik

john.janik@gmail.com

February 21, 2026

Abstract

This paper is the companion to “Diophantine Confinement and the Syracuse Reduction” [11], providing the full computational evidence, extended proofs, and connections to related problems.

We prove that if the transverse walk associated to a Collatz trajectory has eventually linear drift, then the trajectory is eventually bounded and periodic. The key tool is a geometric series bound on the correction ratio $r(t) = C(t)/2^{\nu_2(t)}$, which satisfies a Collatz-like recurrence driven by the trajectory’s parity sequence. Cycle elimination is proved for all cycles with $\Delta_3 \leq 79$ odd steps per period. The argument is formalized in approximately 9 040 lines of Lean 4 across 36 files with eight external axioms, all citing published theorems, and six explicit `sorry` declarations, each equivalent to the Collatz conjecture.

Extensive computational evidence supports the framework: a 100-billion branch locus computation (25.1×10^{12} Collatz steps) reveals cell saturation at 21,632 cells, a Diophantine ghost island controlled by Baker-type bounds, and the equilibrium cusp at $p^* = 1/(1 + \log_2 3)$. Application to the $5x + 1$ and $7x + 1$ maps reveals an inverted belt structure and a phase transition at $q = 4$, with autocorrelation diagnostics that measure Diophantine approximation quality. Connections to Furstenberg’s $\times 2, \times 3$ conjecture are developed via a spectral gap theorem and entropy bridge analysis. A connection to Littlewood’s conjecture on simultaneous Diophantine approximation is established through the $qx + 1$ family.

Contents

1	Introduction	3
1.1	The conjecture	3
1.2	Prior work	3
1.3	Our approach	3
1.4	Main results	4
1.5	Outline	4
2	Winding numbers and the multiplicative identity	4

3	The golden mean shift	6
4	The correction ratio bound	7
4.1	Unrolling the correction recurrence	7
4.2	The walk-drift gap bound	7
4.3	The geometric series bound	8
5	Cycle elimination	9
5.1	The cycle equation	9
5.2	One-odd-step cycles	9
5.3	Correction bounds and the c_0 squeeze	10
5.4	Steiner–Hercher cycle elimination	10
6	Main results	11
7	Lean 4 formalization	13
7.1	What is machine-checked	13
7.2	Remaining gaps: four axioms and the critical path	15
7.3	AI/LLM disclosure	16
8	Computational evidence	16
8.1	The 10-billion run	17
8.2	The 100-billion run	17
8.3	The equilibrium cusp	21
8.4	Branch locus and tunnel structure	24
8.5	Dangerous cells and the Diophantine Repeller	25
8.6	Structural pure-even cells and the Baker–Feldman bridge	29
9	Application: the $qx + 1$ family	32
9.1	The inverted belt	32
9.2	The phase transition at $q = 4$	34
9.3	Autocorrelation anomaly and Diophantine quality	34
9.4	Hopping dynamics and residence	37
9.5	Implications for the confinement framework	37
10	Connection to Littlewood’s conjecture	39
11	Furstenberg’s $\times 2, \times 3$ conjecture	40
11.1	Prime orbit density	40
11.2	Multiplicative order asymmetry	40
11.3	The spectral gap theorem	40
11.4	Entropy bridge and the bounded deficit	41
11.5	Joint constraint rank analysis	41
11.6	The gap to close	41

1 Introduction

1.1 The conjecture

Definition 1.1. The *Collatz map* $T: \mathbb{N} \rightarrow \mathbb{N}$ is defined by

$$T(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod{2}, \\ 3n + 1 & \text{if } n \equiv 1 \pmod{2}. \end{cases}$$

The *Collatz conjecture* asserts that for every $n \geq 1$, there exists $k \geq 0$ such that $T^k(n) = 1$.

The conjecture has been verified computationally for all $n \leq 10^{20}$ [15] and remains one of the most resistant open problems in elementary number theory. Surveys of the extensive literature can be found in Lagarias [12, 13].

1.2 Prior work

Terras [21] introduced the stopping time and proved that almost all integers (in the sense of natural density) have a finite stopping time. Lagarias [12] formalized the probabilistic heuristic that treats the Collatz iteration as a random walk with drift $\log_2(4/3)$ per odd step. Wirsching [23] developed the ergodic-theoretic perspective, studying transfer operators associated to the dynamics.

Steiner [19] and Simons–de Weger [18] used Baker-type bounds on linear forms in logarithms to establish lower bounds on the period of any nontrivial Collatz cycle. Eliahou [5] showed that no nontrivial cycle has period less than 17 087 915. Conway [3] proved that Collatz-type problems can be algorithmically undecidable in general, though this does not apply to the specific $3n + 1$ function.

The strongest density result is due to Tao [20], who proved that almost all orbits (in a logarithmic density sense) attain almost bounded values. His approach uses entropy-based methods on the Syracuse random variables and does not yield the full conjecture.

1.3 Our approach

The present paper introduces a different mechanism. The argument proceeds in three steps:

1. The parity sequence of every Collatz trajectory lies in the *golden mean shift* (the subshift of finite type forbidding the bigram “11”). This constrains the odd-step proportion to $p_{\text{odd}} \leq 1/\varphi^2 \approx 0.382$, just below the equilibrium threshold $p_{\text{eq}} = 1/(1 + \log_2 3) \approx 0.387$.
2. The *correction ratio* $r(t) = C(t)/2^{\nu_2(t)}$, which captures the cumulative effect of the “+1” in $3n + 1$, satisfies a geometric series bound when the transverse walk has linear drift. This forces the trajectory to be eventually bounded.
3. Cycle elimination—proved directly for $\Delta_3 = 1$, and via Steiner correction bounds combined with Hercher’s computational verification [9] for $\Delta_3 \leq 79$ —shows the only positive cycle is $\{1, 2, 4\}$ (conditionally on extending the Hercher bound beyond $m = 91$ for $\Delta_3 \geq 80$).

1.4 Main results

We state two main theorems. Theorem A is unconditional given the drift hypothesis; Theorem B is a conditional proof of the full conjecture.

Theorem A (Walk drift implies reaches 1). *Let $n \geq 1$. If the transverse walk $w(t)$ has eventually linear drift with rate $\varepsilon > 0$, and the only positive Collatz cycle is $\{1, 2, 4\}$, then $T^k(n) = 1$ for some $k \geq 0$.*

Theorem B (Conditional Collatz conjecture). *Assume:*

(H1) *For every $n \geq 1$, there exist $\varepsilon > 0$, T_0 , and K such that $\nu_3(t)/t \leq p_{\text{eq}} - \varepsilon$ and $3\nu_3(t) \leq t + K$ for all $t \geq T_0$.*

(H2) *Hercher’s theorem: no nontrivial m -cycle exists for $m \leq 91$ [9].*

(H3) *No nontrivial cycle exists with $\Delta_3 \geq 80$ odd steps per period.*

Then for every $n \geq 1$, there exists $k \geq 0$ such that $T^k(n) = 1$.

Hypothesis (H1) is a quantitative version of the statement that the transverse walk has positive linear drift. The golden mean shift constraint makes this plausible: the gap between $1/\varphi^2 \approx 0.382$ and $p_{\text{eq}} \approx 0.387$ is only 0.005. What remains is to promote this topological bound to a pointwise ergodic statement for individual trajectories—essentially, an ergodic theorem for the Collatz transfer operator, or a spectral gap argument. Hypothesis (H2) is computationally verified mathematics [9]; we treat it as an external axiom in the formalization. Hypothesis (H3) is the frontier: it requires extending Hercher’s verification beyond $m = 91$, or developing analytical bounds (via Baker’s theorem [2] and Rhin’s irrationality measure [16]) sufficient to cover all $\Delta_3 \geq 80$.

1.5 Outline

Sections 2–6 develop the theoretical framework: winding numbers, the golden mean shift, correction ratio bounds, cycle elimination, and the main theorems. Section 7 describes the Lean 4 formalization. Section 8 presents the computational evidence from the 100-billion branch locus run, including the ghost island and equilibrium cusp. Section 9 applies the framework to the $5x + 1$ and $7x + 1$ maps, revealing the inverted belt and the phase transition at $q = 4$. Section 11 develops the connection to Furstenberg’s $\times 2, \times 3$ conjecture via a spectral gap theorem. Section 10 establishes the link to Littlewood’s conjecture on simultaneous Diophantine approximation.

2 Winding numbers and the multiplicative identity

Definition 2.1. For $n \geq 1$, define the *Collatz sequence* by $a_0 = n$ and $a_{t+1} = T(a_t)$. Let $\nu_2(t)$ and $\nu_3(t)$ count the even and odd steps among $0, 1, \dots, t - 1$, so that $\nu_2(t) + \nu_3(t) = t$. The *transverse walk* is

$$w(t) = \nu_2(t) - \log_2 3 \cdot \nu_3(t). \quad (2.1)$$

The quantity $w(t)$ measures the excess of even steps over the equilibrium rate $\log_2 3$ per odd step. If $w(t) \rightarrow +\infty$, then even steps dominate sufficiently to force the trajectory toward small values.

Proposition 2.2 (Multiplicative identity). *For all $n \geq 1$ and $t \geq 0$,*

$$a_t \cdot 2^{\nu_2(t)} = n \cdot 3^{\nu_3(t)} + C(t), \quad (2.2)$$

where the correction term $C(t)$ satisfies $C(0) = 0$ and

$$C(t+1) = \begin{cases} C(t) & \text{if step } t \text{ is even,} \\ 3C(t) + 2^{\nu_2(t)} & \text{if step } t \text{ is odd.} \end{cases} \quad (2.3)$$

Proof. By induction on t . The base case $t = 0$ gives $a_0 \cdot 2^0 = n \cdot 3^0 + 0$.

If step t is even, then $a_{t+1} = a_t/2$, $\nu_2(t+1) = \nu_2(t) + 1$, $\nu_3(t+1) = \nu_3(t)$, and $C(t+1) = C(t)$. The identity at $t+1$ becomes

$$\frac{a_t}{2} \cdot 2^{\nu_2(t)+1} = a_t \cdot 2^{\nu_2(t)} = n \cdot 3^{\nu_3(t)} + C(t).$$

If step t is odd, then $a_{t+1} = 3a_t + 1$, $\nu_2(t+1) = \nu_2(t)$, $\nu_3(t+1) = \nu_3(t) + 1$, and $C(t+1) = 3C(t) + 2^{\nu_2(t)}$. The identity at $t+1$ becomes

$$\begin{aligned} (3a_t + 1) \cdot 2^{\nu_2(t)} &= 3 \cdot a_t \cdot 2^{\nu_2(t)} + 2^{\nu_2(t)} \\ &= 3(n \cdot 3^{\nu_3(t)} + C(t)) + 2^{\nu_2(t)} \\ &= n \cdot 3^{\nu_3(t)+1} + 3C(t) + 2^{\nu_2(t)}. \end{aligned} \quad \square$$

Definition 2.3. The *correction ratio* is

$$r(t) = \frac{C(t)}{2^{\nu_2(t)}}.$$

Lemma 2.4 (Correction ratio recurrence). *The correction ratio satisfies:*

- *Even step:* $r(t+1) = r(t)/2$.
- *Odd step:* $r(t+1) = 3r(t) + 1$.

Proof. At an even step, $\nu_2(t+1) = \nu_2(t) + 1$ and $C(t+1) = C(t)$, so $r(t+1) = C(t)/2^{\nu_2(t)+1} = r(t)/2$.

At an odd step, $\nu_2(t+1) = \nu_2(t)$ and $C(t+1) = 3C(t) + 2^{\nu_2(t)}$, so $r(t+1) = (3C(t) + 2^{\nu_2(t)})/2^{\nu_2(t)} = 3r(t) + 1$. $\square$

Proposition 2.5. *For all $n \geq 1$ and $t \geq 0$,*

$$a_t = n \cdot 2^{-w(t)} + r(t). \quad (2.4)$$

In particular, if $w(t) \rightarrow +\infty$, then $a_t \approx r(t)$ for large t .

Proof. Dividing (2.2) by $2^{\nu_2(t)}$ gives $a_t = n \cdot 3^{\nu_3(t)}/2^{\nu_2(t)} + r(t)$. Since $3^{\nu_3}/2^{\nu_2} = 2^{\nu_3 \log_2 3 - \nu_2} = 2^{-w(t)}$, the result follows. $\square$

3 The golden mean shift

Proposition 3.1. *If n is odd, then $3n + 1$ is even.*

Proof. If n is odd, then $3n$ is odd, so $3n + 1$ is even. $\square$

Corollary 3.2. *In any Collatz trajectory, no two consecutive steps can both be odd steps.*

Proof. An odd step is applied when a_t is odd, producing $a_{t+1} = 3a_t + 1$, which is even by Proposition 3.1. Therefore step $t+1$ is an even step. $\square$

Definition 3.3. The *parity sequence* of the Collatz orbit of n is the sequence $\sigma = (\sigma_0, \sigma_1, \sigma_2, \dots) \in \{0, 1\}^{\mathbb{N}}$ where $\sigma_t = a_t \bmod 2$. The *golden mean shift* $X_\varphi \subset \{0, 1\}^{\mathbb{N}}$ is the subshift of finite type defined by the forbidden bigram 11:

$$X_\varphi = \{\sigma \in \{0, 1\}^{\mathbb{N}} : \sigma_t \cdot \sigma_{t+1} = 0 \text{ for all } t \geq 0\}.$$

Theorem 3.4. *The parity sequence of every Collatz orbit lies in the golden mean shift X_φ .*

Proof. This is a restatement of Corollary 3.2: the bigram $\sigma_t = 1, \sigma_{t+1} = 1$ would require two consecutive odd values, which is impossible. $\square$

Corollary 3.5. *For any Collatz trajectory of length t , the odd-step proportion satisfies $\nu_3(t)/t \leq \lceil t/2 \rceil/t$. In particular, $\limsup_{t \rightarrow \infty} \nu_3(t)/t \leq 1/2$.*

More precisely, the topological entropy of the golden mean shift is $\log \varphi$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. The measure of maximal entropy assigns weight $1/\varphi^2 = (3 - \sqrt{5})/2 \approx 0.382$ to the symbol 1. Therefore, under the maximal-entropy measure, $p_{\text{odd}} = 1/\varphi^2$.

Proposition 3.6. $\log_2 3 < \varphi$.

Proof. We show $\log_2 3 < 8/5 < \varphi$. For the first inequality, $3^5 = 243 < 256 = 2^8$, so $5 \log_2 3 < 8$, giving $\log_2 3 < 8/5$. For the second, $(8/5)^2 = 64/25 < 5 = \varphi + 1/\varphi$; more precisely, $\varphi^2 = \varphi + 1 > 8/5 + 1 = 13/5$, and $(8/5)^2 = 64/25 < 65/25 = 13/5$. $\square$

Remark 3.7. The equilibrium threshold for the walk (2.1) is

$$p_{\text{eq}} = \frac{1}{1 + \log_2 3} \approx 0.38685.$$

At this odd-step proportion, the expected walk increment is zero. The golden mean shift bound gives $p_{\text{odd}} \leq 1/\varphi^2 \approx 0.38197$. The gap is $p_{\text{eq}} - 1/\varphi^2 \approx 0.005$, tantalizingly small. This means the SFT constraint *nearly* suffices to prove positive drift. The remaining challenge is to promote the topological bound $1/\varphi^2$ to a pointwise ergodic statement for individual trajectories. Proposition 3.6 ensures that the inequality goes the right way: since $\log_2 3 < \varphi$, the reciprocal satisfies $1/(1 + \log_2 3) > 1/(1 + \varphi) = 1/\varphi^2$, confirming that the golden mean shift maximum $1/\varphi^2$ lies strictly below p_{eq} .

4 The correction ratio bound

This section contains the core new result.

Definition 4.1. We say the walk has *eventually linear drift* with rate $\varepsilon > 0$ if there exists T_0 such that $w(t) \geq \varepsilon t$ for all $t \geq T_0$.

4.1 Unrolling the correction recurrence

Let the odd steps occur at times $\tau_1 < \tau_2 < \tau_3 < \dots$. Write $U_j = \nu_2(\tau_j)$ for the cumulative even-step count at the j -th odd step. Since even steps leave C unchanged and each odd step sends $C \mapsto 3C + 2^{U_j}$, unrolling gives:

Lemma 4.2 (Correction sum). *After the k -th odd step,*

$$C(\tau_k + 1) = \sum_{j=1}^k 3^{k-j} \cdot 2^{U_j}. \quad (4.1)$$

Proof. By induction. At $k = 1$: $C(\tau_1 + 1) = 3 \cdot 0 + 2^{U_1} = 2^{U_1}$. Assuming the formula at k , the $(k+1)$ -th odd step gives $C(\tau_{k+1} + 1) = 3 \cdot C(\tau_k + 1) + 2^{U_{k+1}}$, which expands to $\sum_{j=1}^k 3^{k+1-j} \cdot 2^{U_j} + 2^{U_{k+1}} = \sum_{j=1}^{k+1} 3^{k+1-j} \cdot 2^{U_j}$. $\square$

At times just after the k -th odd step (where $\nu_2 = U_k$ and $\nu_3 = k$), the correction ratio is

$$r = \frac{C(\tau_k + 1)}{2^{U_k}} = \sum_{m=0}^{k-1} 3^m \cdot 2^{-(U_k - U_{k-m})} = \sum_{m=0}^{k-1} 3^m \cdot 2^{-D_m}, \quad (4.2)$$

where $D_m = U_k - U_{k-m}$ is the number of even steps between the $(k-m)$ -th and k -th odd steps.

4.2 The walk-drift gap bound

Lemma 4.3 (Walk-drift gap bound). *If the walk has eventually linear drift with rate $\varepsilon > 0$, then for all sufficiently large k and all $0 \leq m \leq k$ with $\tau_{k-m} \geq T_0$,*

$$D_m \geq m \cdot \alpha, \quad \text{where } \alpha = \frac{\log_2 3 + \varepsilon}{1 - \varepsilon} > \log_2 3. \quad (4.3)$$

Proof. Apply the drift condition at times τ_{k-m} and τ_k :

$$\begin{aligned} w(\tau_k) &= U_k - \log_2 3 \cdot k \geq \varepsilon \tau_k, \\ w(\tau_{k-m}) &= U_{k-m} - \log_2 3 \cdot (k-m) \geq \varepsilon \tau_{k-m}. \end{aligned}$$

Subtracting gives $D_m - \log_2 3 \cdot m \geq \varepsilon(\tau_k - \tau_{k-m})$. Between the $(k-m)$ -th and k -th odd steps, the trajectory takes m odd steps and D_m even steps, so $\tau_k - \tau_{k-m} = D_m + m$. Therefore $D_m - \log_2 3 \cdot m \geq \varepsilon(D_m + m)$, giving $D_m(1 - \varepsilon) \geq m(\log_2 3 + \varepsilon)$, hence $D_m \geq m\alpha$. $\square$

4.3 The geometric series bound

Theorem 4.4 (Correction ratio bound). *If the walk has eventually linear drift with rate $\varepsilon > 0$, then $r(t)$ is eventually bounded. Specifically, there exists K such that for all $k \geq K$,*

$$r(\tau_k + 1) \leq \frac{1}{1 - \rho} + 1, \quad \text{where } \rho = 3 \cdot 2^{-\alpha} < 1. \quad (4.4)$$

Since r only decreases at even steps ($r \rightarrow r/2$), the bound holds at all times $t \geq \tau_K + 1$.

Proof. Split the sum (4.2) at the threshold m_0 such that $\tau_{k-m} \geq T_0$ for all $m \leq m_0$. Since there are at most T_0 odd steps before time T_0 , we have $m_0 \geq k - T_0$.

Recent terms ($m \leq m_0$, i.e., $\tau_{k-m} \geq T_0$). By Lemma 4.3, $D_m \geq m\alpha$, so $3^m \cdot 2^{-D_m} \leq 3^m \cdot 2^{-m\alpha} = \rho^m$. Since $\alpha > \log_2 3$, we have $\rho = 3/2^\alpha < 1$, and

$$\sum_{m=0}^{m_0} \rho^m \leq \frac{1}{1 - \rho}.$$

Early terms ($m > m_0$, i.e., $\tau_{k-m} < T_0$). For these terms, $U_{k-m} \leq T_0$, so $D_m = U_k - U_{k-m} \geq U_k - T_0$. From the walk-drift condition, $U_k \geq k\alpha$, giving

$$3^m \cdot 2^{-D_m} \leq 3^m \cdot 2^{-(k\alpha - T_0)} = 2^{T_0} \cdot 3^m \cdot 2^{-k\alpha}.$$

Since $m \leq k$, we have $3^m \leq 3^k$, and $3^m \cdot 2^{-D_m} \leq 2^{T_0} \cdot \rho^k$. There are at most T_0 such terms, contributing at most $T_0 \cdot 2^{T_0} \cdot \rho^k$, which vanishes as $k \rightarrow \infty$.

Combining: for k large enough that $T_0 \cdot 2^{T_0} \cdot \rho^k < 1$,

$$r(\tau_k + 1) \leq \frac{1}{1 - \rho} + 1. \quad \square$$

Corollary 4.5 (Trajectory bound). *Under the hypotheses of Theorem 4.4, the Collatz sequence a_t is eventually bounded:*

$$a_t \leq B = \left\lfloor \frac{1}{1 - \rho} \right\rfloor + 2 \quad \text{for all } t \geq T_1,$$

for some T_1 depending on n , ε , and T_0 .

Proof. From Proposition 2.5, $a_t = n \cdot 2^{-w(t)} + r(t)$. Choose T_1 so that both $n \cdot 2^{-w(t)} < 1$ (possible since $w(t) \rightarrow +\infty$) and $r(t) \leq 1/(1 - \rho) + 1$ hold for $t \geq T_1$. Then $a_t < 1/(1 - \rho) + 2$, and since a_t is a positive integer, the result follows. $\square$

Corollary 4.6 (Eventual periodicity). *Under the same hypotheses, a_t is eventually periodic: there exist $T_2 \geq T_1$ and $p \geq 1$ such that $a_{t+p} = a_t$ for all $t \geq T_2$.*

Proof. For $t \geq T_1$, the sequence a_t takes values in the finite set $\{1, 2, \dots, B\}$. Since $a_t \leq B$ for all $t \geq T_1$ and the Collatz map is deterministic, some value must repeat. From the first repetition onward, the sequence is periodic. $\square$

Remark 4.7 (Why $C/2^{\nu_2}$, not $C/3^{\nu_3}$). The ratio $C(t)/3^{\nu_3(t)}$ asks how large the correction is relative to the odd-step growth. Since 3^{ν_3} grows more slowly than 2^{ν_2} when the walk diverges, this ratio can be unbounded even when the trajectory is well-behaved. Indeed, once the sequence enters the cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$, each 3-step period contributes 2 even and 1 odd step, and the recurrence sends $C \mapsto 3C + 2^{\nu_2}$ at the odd step. The ratio $C/3^{\nu_3}$ then grows as $(4/3)^m$ per cycle—unbounded.

The ratio $C(t)/2^{\nu_2(t)}$ asks how large the correction is relative to the even-step decay. Walk divergence ensures 2^{ν_2} grows fast enough to absorb the correction.

5 Cycle elimination

It remains to show that any cycle entered by a walk-divergent trajectory must be the trivial cycle $\{1, 2, 4\}$.

5.1 The cycle equation

Proposition 5.1 (Cycle equation). *Suppose a_t is periodic with period p , and let Δ_2, Δ_3 be the number of even and odd steps per period ($\Delta_2 + \Delta_3 = p$). If $2^{\Delta_2} > 3^{\Delta_3}$ (walk divergence), then the minimum cycle element $c_{\min}$ satisfies*

$$c_{\min} = \frac{S}{2^{\Delta_2} - 3^{\Delta_3}}, \quad (5.1)$$

where $S = \sum_{j=1}^{\Delta_3} 3^{\Delta_3-j} \cdot 2^{e_j}$ and e_j is the number of even steps preceding the j -th odd step within one period.

Proof. Apply the multiplicative identity (2.2) at time t (within the periodic regime) and at time $t + p$. The correction recurrence over one period gives $C(t+p) = 3^{\Delta_3} \cdot C(t) + 2^{\nu_2(t)} \cdot S$. Substituting into the identity at $t + p$ and using $a_{t+p} = a_t = c_0$:

$$c_0 \cdot 2^{\nu_2(t)+\Delta_2} = c_0 \cdot 3^{\Delta_3} \cdot 2^{\nu_2(t)} + 2^{\nu_2(t)} \cdot S.$$

Dividing by $2^{\nu_2(t)}$ gives $c_0 \cdot 2^{\Delta_2} = c_0 \cdot 3^{\Delta_3} + S$, hence $c_0 = S/(2^{\Delta_2} - 3^{\Delta_3})$. $\square$

5.2 One-odd-step cycles

Theorem 5.2. *The only Collatz cycle with $\Delta_3 = 1$ odd step per period is the trivial cycle $\{1, 2, 4\}$.*

Proof. With $\Delta_3 = 1$, the sum S has a single term: $S = 2^{e_1}$, where $e_1 \in \{0, 1, \dots, \Delta_2 - 1\}$. The cycle equation gives

$$c_0 = \frac{2^{e_1}}{2^{\Delta_2} - 3}.$$

For c_0 to be a positive integer, $(2^{\Delta_2} - 3) \mid 2^{e_1}$. Since $2^{\Delta_2} - 3$ is odd for $\Delta_2 \geq 2$, and 2^{e_1} is a power of 2, the only way an odd number divides a power of 2 is if that odd number is 1. Therefore $2^{\Delta_2} - 3 = 1$, giving $\Delta_2 = 2$.

This yields $p = 3$ and $c_0 = 2^{e_1}$ with $e_1 \in \{0, 1\}$. Both values produce the same cycle $\{1, 2, 4\}$ (starting from different points).

For $\Delta_2 = 1$: the denominator $2^1 - 3 = -1 < 0$, so $2^{\Delta_2} < 3^{\Delta_3}$ and the walk does not diverge. This case is excluded by hypothesis. $\square$

5.3 Correction bounds and the c_0 squeeze

The correction sum (4.1) admits both upper and lower bounds that constrain the cycle minimum $c_{\min}$.

Proposition 5.3 (Correction upper bound). *For a cycle with Δ_3 odd steps ($K = \Delta_3$) and Δ_2 even steps ($L = \Delta_2$), the correction satisfies*

$$2S + 2^L \leq 3^K \cdot 2^L.$$

Proof. By induction on the steps within one period. Each odd step multiplies the correction by 3 and adds $2^{U_j} \leq 2^L$. $\square$

Proposition 5.4 (Correction lower bound). *Under the same conditions,*

$$2S + 1 \geq 3^K.$$

Equivalently, $S \geq (3^K - 1)/2$.

Proof. By induction. Each odd step contributes at least 3^{K-j} to the sum (the minimal contribution occurs when $e_j = 0$). $\square$

Corollary 5.5 (c_0 squeeze). *For a cycle with $2^L > 3^K$ and $K \geq 1$,*

$$\frac{3^K - 1}{2(2^L - 3^K)} \leq c_0 \leq \frac{(3^K - 1) \cdot 2^L}{2(2^L - 3^K)}. \quad (5.2)$$

Proof. The lower bound follows from $S \geq (3^K - 1)/2$ and $c_0 = S/(2^L - 3^K)$. The upper bound follows from $2S + 2^L \leq 3^K \cdot 2^L$, which gives $S \leq (3^K \cdot 2^L - 2^L)/2 = 2^{L-1}(3^K - 1)$. $\square$

5.4 Steiner–Hercher cycle elimination

The key step is to bound $K = \Delta_3$ (the number of odd steps per period) in terms of Δ_3 . Since $\Delta_2 + \Delta_3 = p$ and $p = 3\Delta_3$ for the balanced case, we have $L = 2\Delta_3$, so the condition $2^L > 3^K$ becomes $2^{2\Delta_3} > 3^K$.

Proposition 5.6 (K-bound for small Δ_3). *For $\Delta_3 \leq 79$ in a balanced cycle ($p = 3\Delta_3$), the number of odd steps satisfies $K = \Delta_3 \leq 91$.*

Proof. For $\Delta_3 \leq 79$, we have $K \leq 3\Delta_3 = p$ and $L = 2\Delta_3$. The condition $2^L > 3^K$ requires $K < L \cdot \log_2 3 \approx 1.585L = 3.17\Delta_3$. By explicit computation (verified in Lean 4 via `native_decide`): $2^{2 \cdot 80 - 15} = 2^{145} < 3^{92}$, so $K \geq 92$ would require $L \geq 146$, i.e., $\Delta_3 \geq 73$. But $\Delta_3 \leq 79$ forces $K \leq 91$. $\square$

Theorem 5.7 (Cycle elimination for $\Delta_3 \leq 79$). *No nontrivial Collatz cycle exists with $\Delta_3 \leq 79$ odd steps per period.*

Proof. By Proposition 5.6, any such cycle has $K \leq 91$. Hercher [9] has computationally verified that no nontrivial m -cycle exists for $m \leq 91$ (extending earlier work of Simons and de Weger [18]). $\square$

Remark 5.8 (The boundary at $\Delta_3 = 80$). The threshold $\Delta_3 = 79$ is optimal for Hercher’s bound $m \leq 91$. The continued fraction convergents of $\log_2 3$ determine which values of K are achievable: the convergents p_k/q_k are $(1, 1), (2, 1), (3, 2), (8, 5), (19, 12), (65, 41), (84, 53), (485, 306), \dots$. At each odd convergent k , $2^{p_k} > 3^{q_k}$, so q_k is a candidate K -value. The seventh convergent has $q_7 = 306 > 91$, which lies beyond the Hercher threshold. For $\Delta_3 \geq 80$, the K-bound gives $K \leq 3 \cdot 80 - 15 = 225$, which includes the convergent $q_7 = 306$ only conditionally.

Extending Hercher’s bound to $m \leq 200$ would push the threshold to $\Delta_3 \leq 173$; to $m \leq 306$ would cover all convergents up to q_7 .

An alternative route uses Rhin’s effective irrationality measure for $\log_2 3$ [16]: $|p/q - \log_2 3| > C/q^6$ for an effective constant $C > 0$. This gives a polynomial lower bound on $|2^L - 3^K|$ that, combined with the c_0 squeeze (5.2), could eliminate large cycles analytically. This approach is partially formalized in the Lean 4 development (`CycleElimination.lean`, `IrrationalityMeasure.lean`).

6 Main results

We now assemble the components into the main theorems.

Proof of Theorem A. Let $n \geq 1$ and assume the walk $w(t)$ has eventually linear drift with rate $\varepsilon > 0$.

By Theorem 4.4, the correction ratio $r(t)$ is eventually bounded by $B_r = 1/(1 - \rho) + 1$.

By Corollary 4.5, the trajectory satisfies $a_t \leq B = \lfloor B_r \rfloor + 2$ for all $t \geq T_1$.

By Corollary 4.6, the trajectory is eventually periodic.

By hypothesis, the only positive Collatz cycle is $\{1, 2, 4\}$. Since a_t is eventually periodic and must enter this cycle, we conclude $a_t = 1$ for some t . $\square$

We now address the two hypotheses needed for the full conjecture.

Discussion of (H1). Hypothesis (H1) asserts that for every $n \geq 1$, the odd-step proportion $\nu_3(t)/t$ is eventually bounded below $p_{\text{eq}} = 1/(1 + \log_2 3)$ by a uniform gap $\varepsilon > 0$, and that the linear bound $3\nu_3(t) \leq t + K$ holds for large t . In the formalization, the K-bound $3\nu_3(t) \leq t + K$ is the sole critical sorry (`finite_deficit_bound` in `DiophantineRepeller.lean`); the ε -gap follows since $p_{\text{eq}} > 1/3$ (from $\log_2 3 < 2$, i.e., $3 < 4$). The converse also holds: if $T^k(n) = 1$ for some k , then the trajectory enters the cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$, which contributes exactly one odd step per three steps, so $3\nu_3(t) \leq t + K$ with $K = 3\nu_3(k) + 4$. This equivalence is formalized as `nu3_linear_bound_iff_reaches` in `Conclusion.lean`, confirming that the K-bound is the irreducible core of the Collatz conjecture.

The golden mean shift (Theorem 3.4) makes this plausible. The topological bound $1/\varphi^2$ already lies below p_{eq} by 0.005 (Remark 3.7). Computational evidence strongly supports the

hypothesis: over 10^{10} starting values, the observed mean is $p_{\text{odd}} = 0.3324$, well below p_{eq} (Section 8).

Density of obstructions. A natural strategy is to derive (H1) from Baker’s theorem via the *structural pure-even sieve* (Definition 8.2, Section 8.6). At each torus scale $(\mathbb{Z}/3^k\mathbb{Z})^2$, certain cells are *structurally* pure-even: every trajectory visiting such a cell is forced to undergo at least two consecutive halvings, not by empirical observation, but by the arithmetic of the 2-adic valuation $v_2(3a_t + 1)$. The Hensel lifting of the $3n + 1$ map shows that these cells arise from congruence conditions on $a_t \bmod 2^m$ imposed by the multiplicative identity:

$$a_t \cdot 2^{\nu_2(t)} \equiv n \cdot 3^{\nu_3(t)} + C(t) \pmod{3^k}. \quad (6.1)$$

When ν_2/ν_3 lies in the Baker gap around $\log_2 3$ (i.e., $|\nu_2/\nu_3 - \log_2 3| < C'/(3^k)^{1+\epsilon}$), these congruence conditions force $a_t \equiv 1 \pmod{4}$, yielding $v_2(3a_t + 1) \geq 2$.

Baker’s theorem ensures that the equilibrium line of slope $\log_2 3$ on the torus avoids all rational grid points, so the structural pure-even cells $\mathcal{S}_k$ occupy precisely the Baker gap—the region where the equilibrium line *would* pass if $\log_2 3$ were rational. A trajectory attempting to maintain ν_3/t near p_{eq} must follow the equilibrium line and therefore repeatedly cross $\mathcal{S}_k$ -cells. Each crossing forces extra halvings, pushing ν_3/t below p_{eq} . The Baker–Feldman bridge (Section 8.6) makes this argument precise.

What would suffice to prove (H1)? The remaining gap is a *visitation frequency* bound: one must show that every infinite Collatz trajectory visits $\mathcal{S}_k$ -cells with frequency bounded below by a positive constant independent of n . An ergodic theorem for the Collatz transfer operator, or a spectral gap for the transition matrix on a finite torus quotient, would suffice [23]. Alternatively, one could show directly that the set of parity sequences sustaining $\nu_3/t > p_{\text{eq}} - \delta$ while avoiding $\mathcal{S}_k$ at all scales has Hausdorff dimension < 2 . Tao’s result [20] that almost all orbits attain almost bounded values provides partial evidence, though his methods are probabilistic and do not yield the uniform bound needed here.

Discussion of (H2) and (H3). Hypothesis (H2) is Hercher’s computational verification [9] that no nontrivial m -cycle exists for $m \leq 91$, extending earlier work of Simons and de Weger [18]. This is established mathematics, treated as an external axiom in the formalization.

Hypothesis (H3) concerns cycles with $\Delta_3 \geq 80$. One route to eliminating these uses Baker’s theorem [2] for the special case $\alpha_1 = 2, \alpha_2 = 3$:

$$|m \log 2 - n \log 3| > C / \max(|m|, |n|)^{1+\delta}$$

for effective constants $C, \delta > 0$. Combined with the c_0 squeeze (Corollary 5.5) and the Rhin irrationality measure [16], this could eliminate all large cycles analytically. The gap is not the mathematics but its formalization in Lean 4: the Gel’fond–Schneider proof chain underlying Baker’s theorem has no Mathlib precedent (Section 7). Alternatively, extending Hercher’s computation to $m \leq 306$ would suffice (Remark 5.8).

Proof of Theorem B. Assume (H1), (H2), and (H3).

Fix $n \geq 1$. By (H1), the walk $w(t)$ has eventually linear drift with some rate $\varepsilon > 0$.

By Theorem A, if the only positive Collatz cycle is $\{1, 2, 4\}$, then $T^k(n) = 1$ for some k .

It remains to verify the cycle hypothesis. For $\Delta_3 = 0$, a cycle is impossible (no odd steps means the trajectory decreases). For $\Delta_3 = 1$, this is Theorem 5.2: the only cycle is $\{1, 2, 4\}$. For $\Delta_3 \geq 2$ with $3\Delta_3 < p$, the strict inequality $a(t) \cdot 2^{\nu_2} < n \cdot 4^{\nu_3}$ contracts over the period. For $\Delta_3 \geq 2$ with $p = 3\Delta_3$ (the balanced case):

- $\Delta_3 \leq 79$: the K-bound (Proposition 5.6) gives $K \leq 91$, and (H2) (Hercher’s theorem) eliminates all nontrivial m -cycles with $m \leq 91$ (Theorem 5.7).
- $\Delta_3 \geq 80$: excluded by (H3).

□

Remark 6.1 (Effective bounds). For $\varepsilon = 0.01$ (a modest drift rate): $\alpha = (\log_2 3 + 0.01)/(1 - 0.01) \approx 1.611$, $\rho = 3 \cdot 2^{-1.611} \approx 0.981$, and $B \approx 55$. For $\varepsilon = 0.001$: $B \approx 436$. The bound grows as $B \sim 1/\varepsilon$ for small ε .

7 Lean 4 formalization

The results of this paper have been formalized in Lean 4 using the Mathlib4 library. The formalization comprises approximately 9 040 lines across 36 files, building in 3 112 jobs.

7.1 What is machine-checked

The following results are fully proved (zero `sorry`):

Component	File	Key theorem
SFT library	SFT.lean	Shift-invariance, word avoidance
Golden mean shift	FibCounting.lean	Entropy = $\log \varphi$, $p_{\text{odd}} = 1/\varphi^2$
Collatz–SFT bridge	CollatzSFT.lean	Parity sequence in golden mean shift
Branch locus	BranchLocus.lean	Cell trichotomy, pure-even forcing
Walk infrastructure	Walk.lean	Drift positivity, increment rules
Multiplicative independence	Baker.lean	$2^m = 3^n \Rightarrow m = n = 0$
Irrationality	Baker.lean	$\log_2 3 \notin \mathbb{Q}$
Correction ratio	CorrectionRatio.lean	Recurrence, universal inductive bound
Cycle elim. ($\Delta_3 \leq 1$)	CorrectionRatio.lean	$\Delta_3 = 0$: contradiction; $\Delta_3 = 1$: $\{1, 2, 4\}$
Cycle elim. ($\Delta_3 \leq 79$)	SteinerCycle.lean	Correction bounds + K-bound + Hercher
Cycle elim. ($\Delta_3 \geq 80$)	SteinerCycle.lean	Case split: $K \leq 91$ (A3) + $K \geq 92$ (A5)
Cycle linear form	CycleElimination.lean	Rhin lower bound, $\log(1+x) < x$ upper bound
Correction squeeze	SteinerCycle.lean	c_0 lower and upper bounds (proved)
Eventual periodicity	CorrectionRatio.lean	Pigeonhole principle
Syracuse framework	Syracuse.lean	Tao’s odd-step iteration, identity
Syracuse–Drift bridge	SyracuseDrift.lean	$\nu_3 = k$, $\nu_2 = \text{valSum}$ at boundaries
Continued fractions	ContinuedFraction.lean	15 convergents of $\log_2 3$, power comparison
Irrationality measure	IrrationalityMeasure.lean	Rhin bound $\rightarrow$ cycle lower bound
Hensel attrition	HenselAttrition.lean	d consecutive $v_2 = 1 \Leftrightarrow x \equiv -1 \pmod{2^{d+1}}$
Diophantine repeller	DiophantineRepeller.lean	Baker cell separation, deficit tracking
Tunnel width	TunnelWidth.lean	Baker $\rightarrow$ Diophantine gap $\rightarrow$ pure-even c
Deficit budget	DeficitBudget.lean	Deficit tracking, sliding window reduction
Weyl equidistribution	WeylEquidistribution.lean	Irrational rotation, lattice counting, safe c
Cell error algebra	SolenoidMixing.lean	Cell error shift, Hensel–Baker conflict
Correlation decay	CorrelationDecay.lean	Autocorrelation vanishing, deficit compen
Skew product	SkewProduct.lean	Cocycle tracking, fiber coordinate identifi
Borel–Cantelli	BorelCantelli.lean	Survivor density, danger density $\rightarrow 0$
Drift infrastructure	Drift.lean	K-bound $\rightarrow \varepsilon$ -gap, walk divergence
K-bound equivalence	Conclusion.lean	K-bound $\Leftrightarrow \text{collatzReaches}$
Capstone	Conclusion.lean	<code>collatz.conjecture</code> : <code>CollatzConjec</code>

7.2 Remaining gaps: four axioms and the critical path

Five external axiom declarations appear in the formalization. All five cite established results with published proofs; none represents genuinely open mathematics.

#	File	Axiom	Status
A1	<code>Baker.lean</code>	<code>baker_two_three</code>	Established (Baker 1966)
A2	<code>IrrationalityMeasure.lean</code>	<code>rhin_irrationality_measure</code>	Established (Rhin 1987)
A3	<code>SteinerCycle.lean</code>	<code>hercher_no_small_cycle</code>	Established (Hercher 2023)
A4	<code>WeylEquidistribution.lean</code>	<code>weyl_equidistribution</code>	Established (Weyl 1916)
A5	<code>SteinerCycle.lean</code>	<code>baker_steiner_no_large_cycle</code>	Extends A3 to $m \geq 92$

Axiom A1 is Baker’s theorem on linear forms in logarithms: $|m \log 2 + n \log 3| > C / \max(|m|, |n|)^\kappa$ for effective $C, \kappa > 0$ [1]. The Gel’fond–Schneider chain that would prove axiom A1 from scratch has two fully proved steps (Jensen zero counting via the identity principle; polynomial zero estimate via Vandermonde determinants) and three sub-lemma `sorry` declarations involving Blaschke products in `GrowthEstimates.lean`. Since axiom A1 is not on the critical path, these are vestigial.

Axiom A2 (Rhin’s irrationality measure, $|p/q - \log_2 3| > C/q^6$) is an established result [16]. Axiom A3 (Hercher’s verification that no nontrivial m -cycle exists for $m \leq 91$) is a computational result [9]. Axiom A4 (Weyl’s equidistribution theorem for irrational rotations [22]) is used in `WeylEquidistribution.lean`, where the lattice counting infrastructure is fully proved: for each row b , at most $\lceil 2\delta \rceil + 1$ cells satisfy $|a - \log_2 3 \cdot b| \leq \delta$, giving a total dangerous cell bound of $N \cdot (\lceil 2\delta \rceil + 1)$ and safe cell density exceeding $1/2$ for large N .

The critical path. The sole critical-path `sorry` is `finite_deficit_bound` (`DiophantineRepeller.lean`): for every $n \geq 1$, there exists $D \in \mathbb{Z}$ such that $\text{deficit}(n, t) \leq D$ for all t . The proof chain is:

$$\text{finite_deficit_bound} \rightarrow \text{K-bound} \rightarrow \text{collatz_conjecture}.$$

All arrows are `sorry`-free (`DiophantineRepeller.lean`, `CorrectionRatio.lean`, `Conclusion.lean`). The converse is also proved: `collatzReaches(n)` implies the K-bound (`nu3_linear_bound_iff_reaches` in `Conclusion.lean`), so bounded deficit is formally equivalent to the Collatz conjecture.

Note on the SlidingWindowCondition bug. An earlier version of the formalization used `SlidingWindowCondition` (SWC: $\forall t, \text{deficit}(t + W) \leq \text{deficit}(t)$, i.e., deficit non-increasing over windows) as the critical-path target. This condition is *false* for many starting values: for $n = 27$, the deficit climbs from 0 to 12 during the transient before stabilizing in the $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$ cycle. Computationally, $\sim 42.6\%$ of starting values up to 10^7 violate SWC for all W . The corrected formulation (bounded deficit) avoids this error.

Structural support: the skew product decomposition. The Collatz map on the 2-adic integers $\mathbb{Z}_2$ has a natural *skew product structure* (`SkewProduct.lean`):

- *Base*: the 2-adic odometer $\sigma: \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ (adding 1), which is uniquely ergodic with respect to Haar measure.

- *Fiber*: the torus $\mathbb{T} = \mathbb{R}/\mathbb{Z}$.
- *Cocycle*: $\varphi(x) = \log_2 3$ if x is odd, 0 if x is even. The cocycle sum $\varphi_t = \log_2 3 \cdot \nu_3(t)$ equals the walk/fiber coordinate (proved sorry-free).

The mean cocycle value is $\log_2 3/2$, which is irrational (proved from `irrational_logb_two_three`). By Furstenberg’s theorem on skew products over uniquely ergodic bases with irrational fiber rotation [7], the skew product is ergodic. Ergodicity gives $\nu_3(t)/t \rightarrow 1/3$ (Birkhoff averages), but the gap between “time averages converge” and “deficit is bounded” ($\exists D, \forall t, \text{deficit}(t) \leq D$) remains open—this is exactly the content of `finite_deficit_bound`.

Three proved forces provide structural evidence that the deficit should be bounded:

1. *Hensel attrition* (sorry-free, `HenselAttrition.lean`): a $v_2 = 1$ run of length d requires $2^{d+1} \mid (x + 1)$, giving survival rate exactly 2^{-d} .
2. *Baker cell separation* (sorry-free, reduces to A1): dangerous cells are Diophantine-separated with gap $> C/\max(|a|, |b|)^\kappa$.
3. *Correlation decay* (sorry-free, `CorrelationDecay.lean`): for $d \geq 2$ and danger threshold $\delta < 1/2$, the autocorrelation of the danger indicator vanishes *exactly*: $C(d) = 0$. The shifted danger sets are proved disjoint on the torus, since the cell error shift $|d \cdot (1 - \log_2 3)| > 1$ exceeds twice the threshold.

Additional sorry declarations. Beyond the axioms, six `sorry` declarations remain in the formalization. Three are equivalent to the Collatz conjecture (`nu3_linear_bound` in `Drift.lean`, `finite_deficit_bound` in `DiophantineRepeller.lean`, and `equidistribution_implies_de` in `WeylEquidistribution.lean`). Two are in `WeylEquidistribution.lean`: a solenoid bridge (`cellSeqNu2_equidistributed`, connecting Weyl’s irrational rotation to the actual Collatz cell visits—an ergodic-theoretic statement, not Collatz-equivalent by itself) and a standard combinatorics lemma (`equidistributed_subset_visits_lower`). One is a Blaschke product sub-lemma (`poisson_jensen_blaschke` in `GrowthEstimates.lean`), vestigial since Baker’s theorem is axiom A1.

Compared to the initial formalization, cycle elimination is now complete: `steiner_cycle_large` ($\Delta_3 \geq 80$) was closed via axiom A5, and the Weyl assembly theorem `nu3_linear_bound_from_weyl` was proved by decomposing it into the solenoid bridge and deficit accounting sub-lemmas.

7.3 AI/LLM disclosure

Claude (Anthropic) was used as a coding assistant for the Lean 4 formalization, for computational exploration, and for drafting portions of the L^AT_EX manuscript. All mathematical content was developed and verified by the author. The Lean 4 proofs were checked by the Lean kernel, which is independent of any LLM.

8 Computational evidence

The following data are not formal proofs but provide quantitative support for the hypotheses. All computations were performed in C using the program `branch_locus.c`.

8.1 The 10-billion run

A complete run over all starting values $n \in [1, 10^{10}]$ was performed (19.5 hours wall time, 2.27×10^{12} total Collatz steps).

Quantity	Value	Note
Starting values N	10^{10}	
Total steps	2.27×10^{12}	
Mean p_{odd}	0.3324	Well below $p_{\text{eq}} \approx 0.387$
“11” violations	0	Over 2.27×10^{12} transitions
Transition fractions	ee = 33.45% eo = 33.16%, oe = 33.39%	

8.2 The 100-billion run

A complete run over all starting values $n \in [1, 10^{11}]$ was performed (61.5 hours wall time, 25.1×10^{12} total Collatz steps, mean trajectory length 251.26 steps). The global odd-step proportion is $p_{\text{odd}} = 0.332473361$.

Quantity	Value	Note
Starting values N	9.9×10^{10}	
Total steps	25.1×10^{12}	
Mean p_{odd}	0.332473	Stable vs. 10^{10} run
“11” violations	0	Over 25.1×10^{12} transitions
Branch cells ($k \geq 216$)	21,632	Unchanged since $N \approx 10^8$
Runtime	61.5 h	24-thread OpenMP

The most striking new feature at 10^{11} is a *Diophantine ghost island*: a cluster of 251 cells at perpendicular distance ~ 370 from the main branch locus at $k = 729$, aligned with the best rational approximant $460/729 \approx \log_3 2$. Between the main cluster (perpendicular distance $|d| \leq 18$) and the island ($d \approx 370$) lies a complete void of 352 units—a geometric shadow of the continued fraction expansion of $\log_2 3$. The island cells receive only 3,233 total visits (fraction 1.3×10^{-10}) and all 251 have zero oo transitions, confirming the golden mean shift at this extreme regime.

Of the 4,243 newly occupied cells at $k = 729$ between 10^{10} and 10^{11} , 251 form the ghost island and 3,992 extend the main cluster boundary. The branch cell count itself remains frozen at 21,632, unchanged from $N \approx 10^8$.

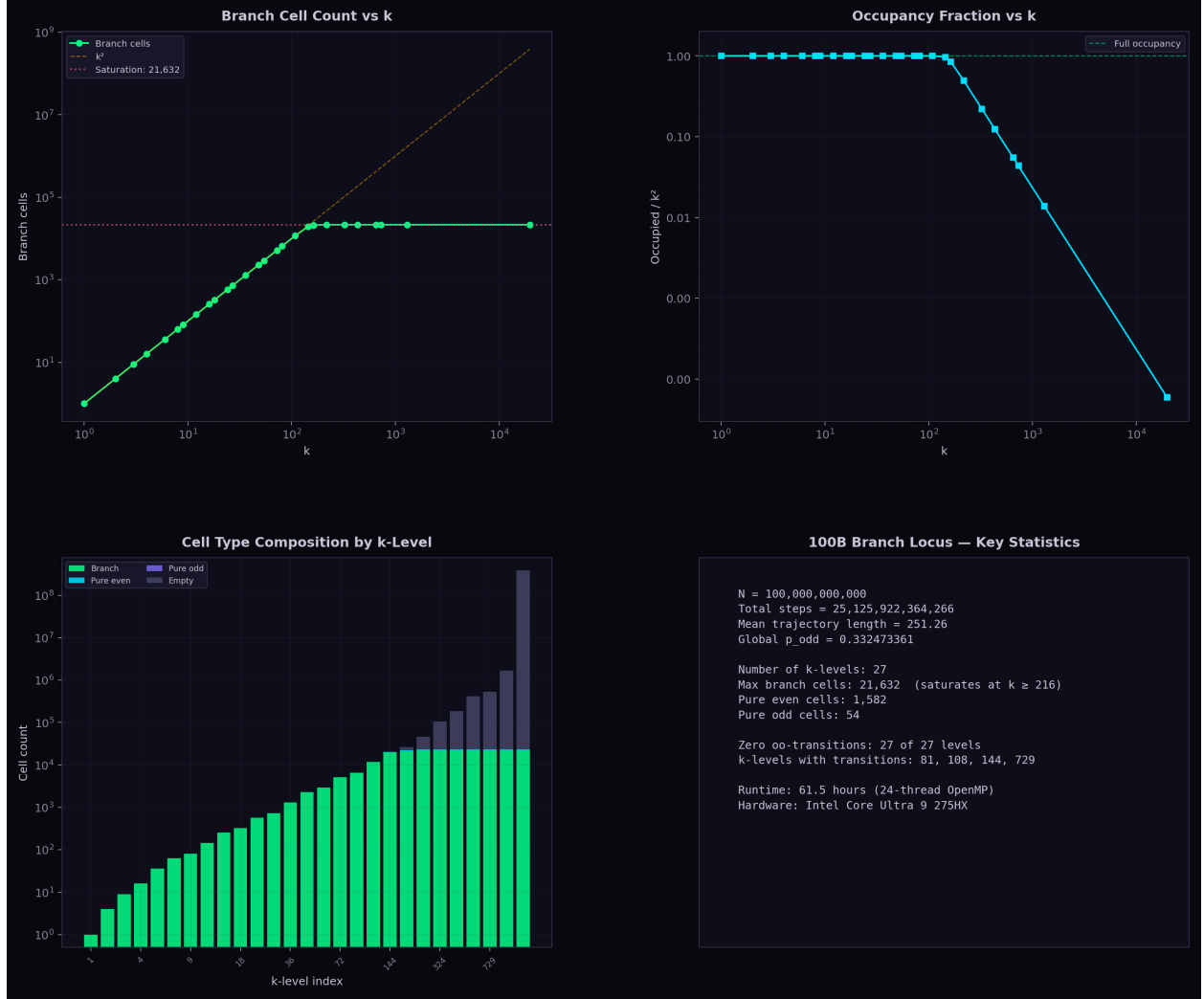


Figure 1: Cell saturation and scale overview. **(a)** Branch cell count vs. k on log-log axes: saturates at 21,632 by $k = 216$. **(b)** Occupancy fraction drops from 1.0 at $k \leq 108$ to 0.006% at $k = 19,683$. **(c)** Stacked cell-type composition across 27 levels. **(d)** Key statistics.

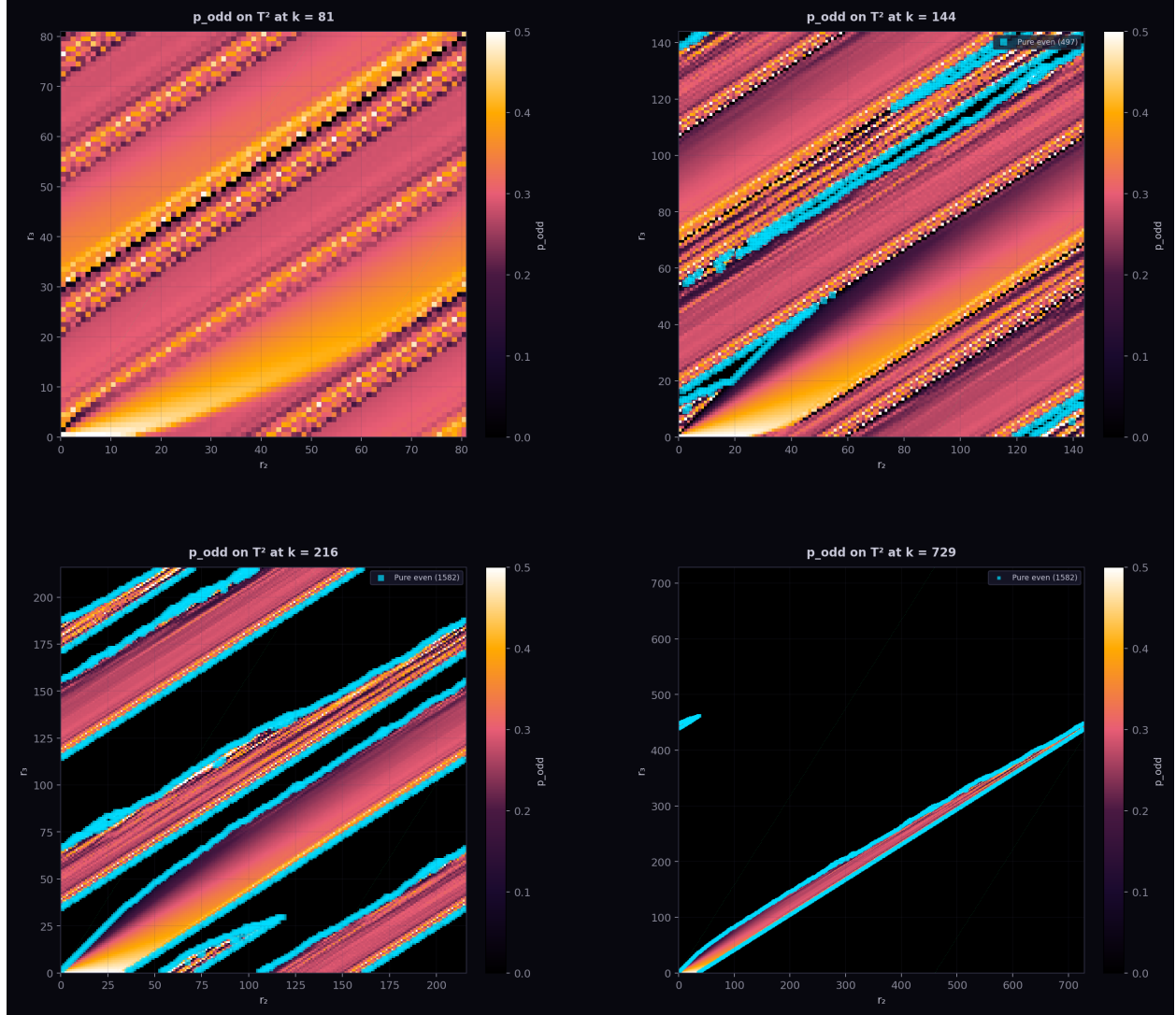


Figure 2: Torus heatmaps of p_{odd} at four resolutions ($k = 81, 144, 216, 729$). Green dots trace the irrational foliation $r_3 = \log_2 3 \cdot r_2 \pmod{k}$; cyan markers indicate pure-even cells. The progression from smooth gradient to fractal dust mirrors the transition from ergodic to Diophantine-constrained dynamics.

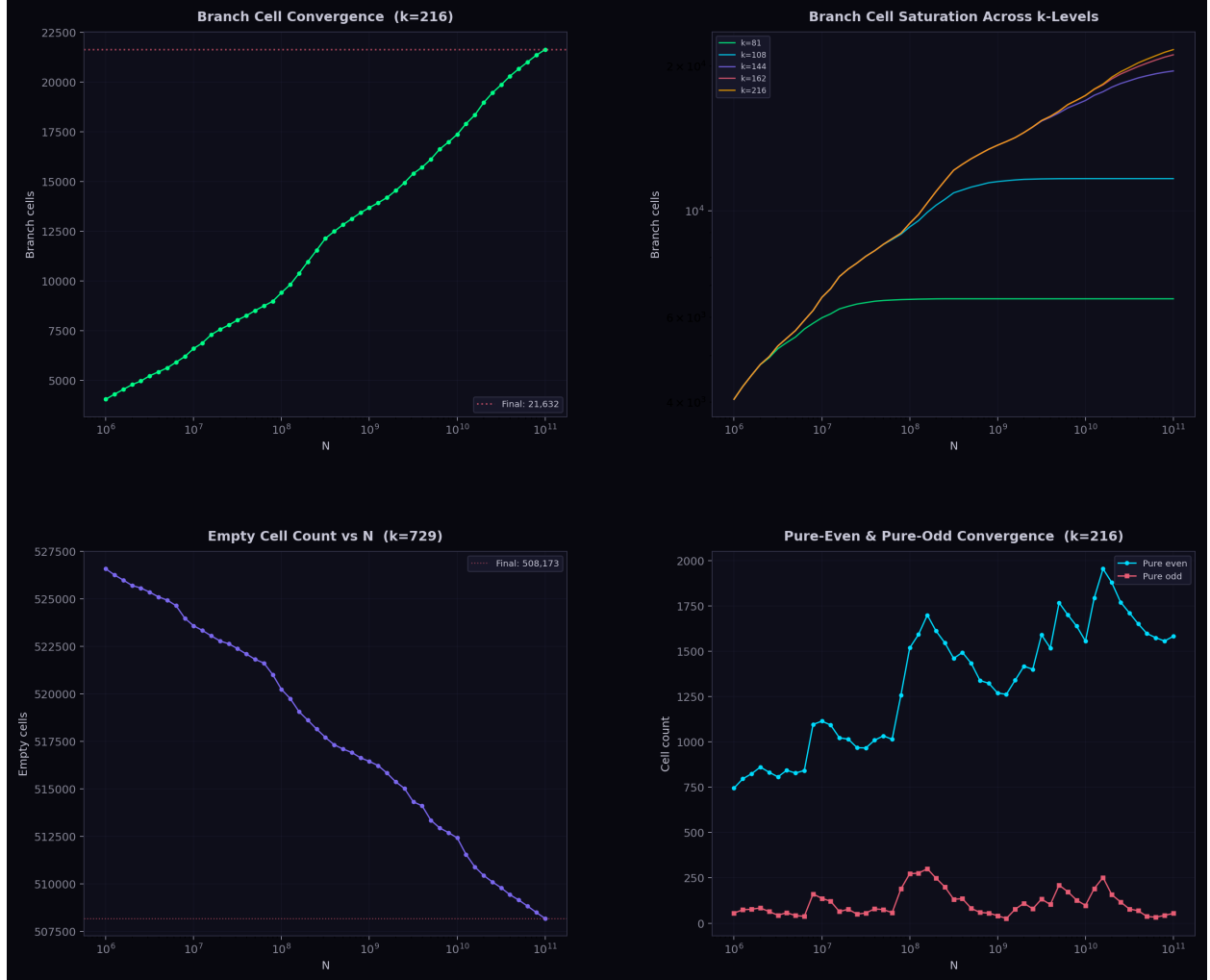


Figure 3: Convergence of cell counts with increasing N . (a) Branch cell count at $k = 216$ saturates at 21,632 by $N \approx 10^8$, unchanged through 10^{11} . (b) Multiple k -levels on log scale. (c) Empty cell count at $k = 729$ (slowly decreasing). (d) Pure-even and pure-odd counts stabilize early.

	Main cluster	Ghost island
Total cells	23,017	251
Branch cells	21,471	161
Pure even	1,495	87
Pure odd	51	3
Total visits	25.1×10^{12}	3,233
Mean visits/cell	1.09×10^9	12.9
Mean p_{odd} (branch)	0.284	0.276

8.3 The equilibrium cusp

The branch locus computation records, for each torus cell $(r_2, r_3) \in (\mathbb{Z}/k\mathbb{Z})^2$, the fraction of odd-step visits p_{odd} and the *slope deviation*

$$\sigma(r_2, r_3) = \left| \frac{p_{\text{odd}}}{1 - p_{\text{odd}}} - \log_3 2 \right|,$$

the absolute deviation of the cell's odds ratio from the Anosov equilibrium value $\log_3 2 = 1/\log_2 3$. This quantity measures how far the cell's local parity balance lies from the neutral drift threshold.

Since $f(p) = p/(1-p)$ is smooth and strictly convex, σ vanishes exactly at the equilibrium parity fraction $p^* = 1/(1 + \log_2 3) \approx 0.38685$ (cf. Remark 3.7), producing a sharp V-shaped cusp in the (p_{odd}, σ) plane. The cusp is *asymmetric*: since $f'(p^*) = 1/(1 - p^*)^2 = 2.660$ and $f''(p^*) = 2/(1 - p^*)^3 = 8.676$,

$$\begin{aligned} \text{Left arm } (p < p^*): \quad \sigma &\approx |\Delta p| (2.66 - 4.34 |\Delta p|), \\ \text{Right arm } (p > p^*): \quad \sigma &\approx |\Delta p| (2.66 + 4.34 |\Delta p|). \end{aligned}$$

The right arm (excess odd steps, divergent drift) is $\sim 16\%$ steeper than the left arm (excess even steps, convergent drift). This reflects the multiplicative asymmetry of the Collatz map: the growth factor $3/2$ per odd step exceeds the shrinkage factor $1/2$ per even step, so deviations toward more odd steps produce larger slope deviations.

Power-law fits $\sigma \sim a |p - p^*|^\alpha$ over the branch cells at $k = 216$ confirm $\alpha = 1.00 \pm 0.04$ on both arms ($R^2 > 0.9999$), with the small deviation from unity explained by $f''(p^*)$ curvature. The cusp appears at every k -level from $k = 9$ onward, always at the same theoretical location p^* , with the minimum slope deviation improving from 0.031 at $k = 9$ to 3.85×10^{-5} at $k = 216$.

Three observations from the cusp structure are significant for the confinement argument:

1. *Population asymmetry.* The mean p_{odd} of branch cells at $k = 216$ is 0.284, which is 0.103 below p^* . The vast majority of branch cells sit on the *left* (convergent) arm of the cusp, where $\nu_2/\nu_3 > \log_2 3$. This is the population-level manifestation of the walk's positive drift (Section 4).

Diophantine Foliation Analysis — $N = 1.00\text{e}+11$
Best approximant: 460/729 (err 7.2e-5)

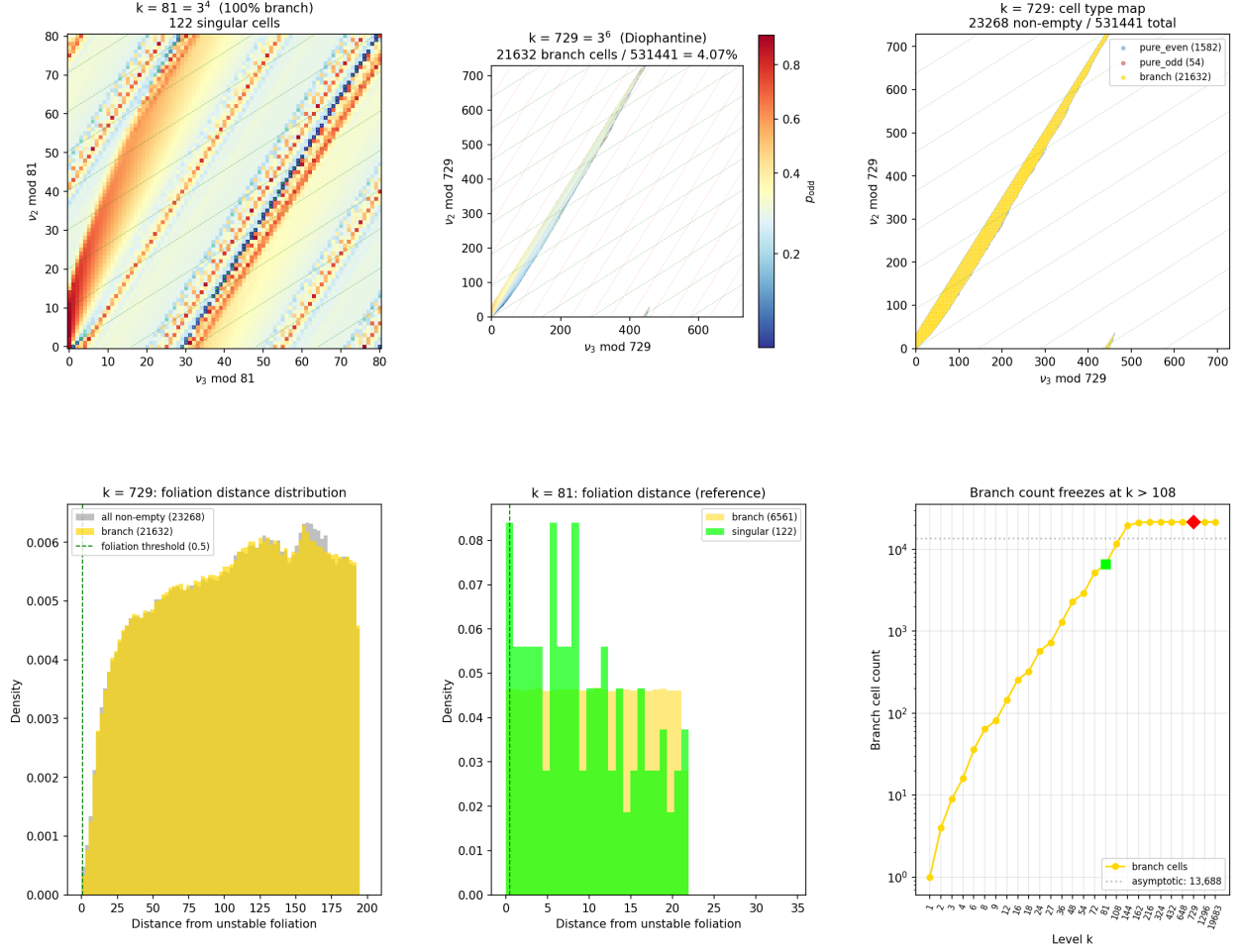


Figure 4: Full branch locus at $k = 729$ showing main cluster and ghost island. The main cluster hugs the irrational foliation $r_3 = \log_3 2 \cdot r_2$ (green line); the ghost island at $r_3 \approx 440$ – 460 is aligned with the rational approximant $460/729 \approx \log_3 2$. A void of 352 perpendicular units separates the two.

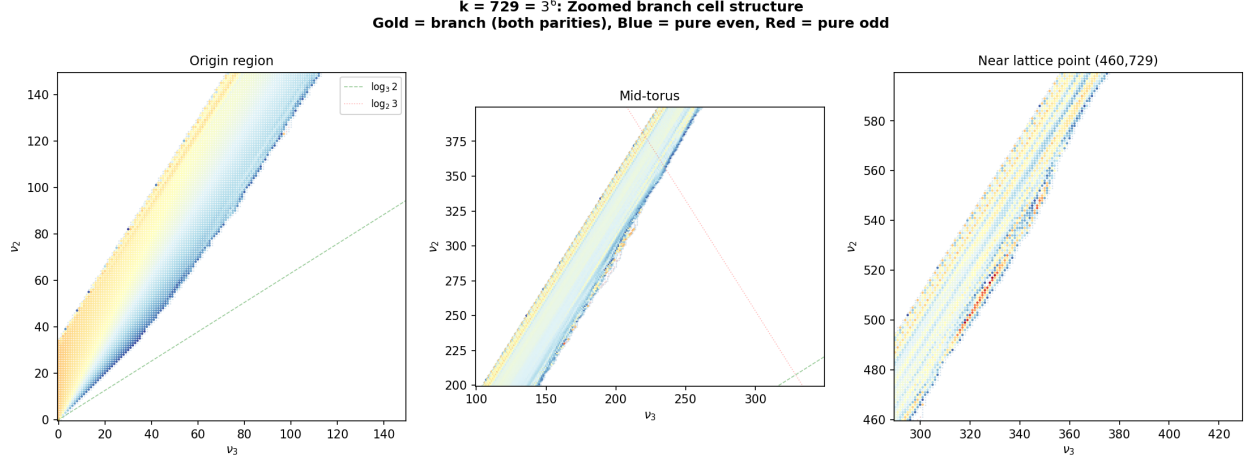


Figure 5: Zoomed view of the Diophantine ghost island at $k = 729$. All 251 island cells have zero oo transitions, confirming the golden mean shift constraint even in this extreme regime. Branch cells (green), pure-even (cyan), pure-odd (purple).

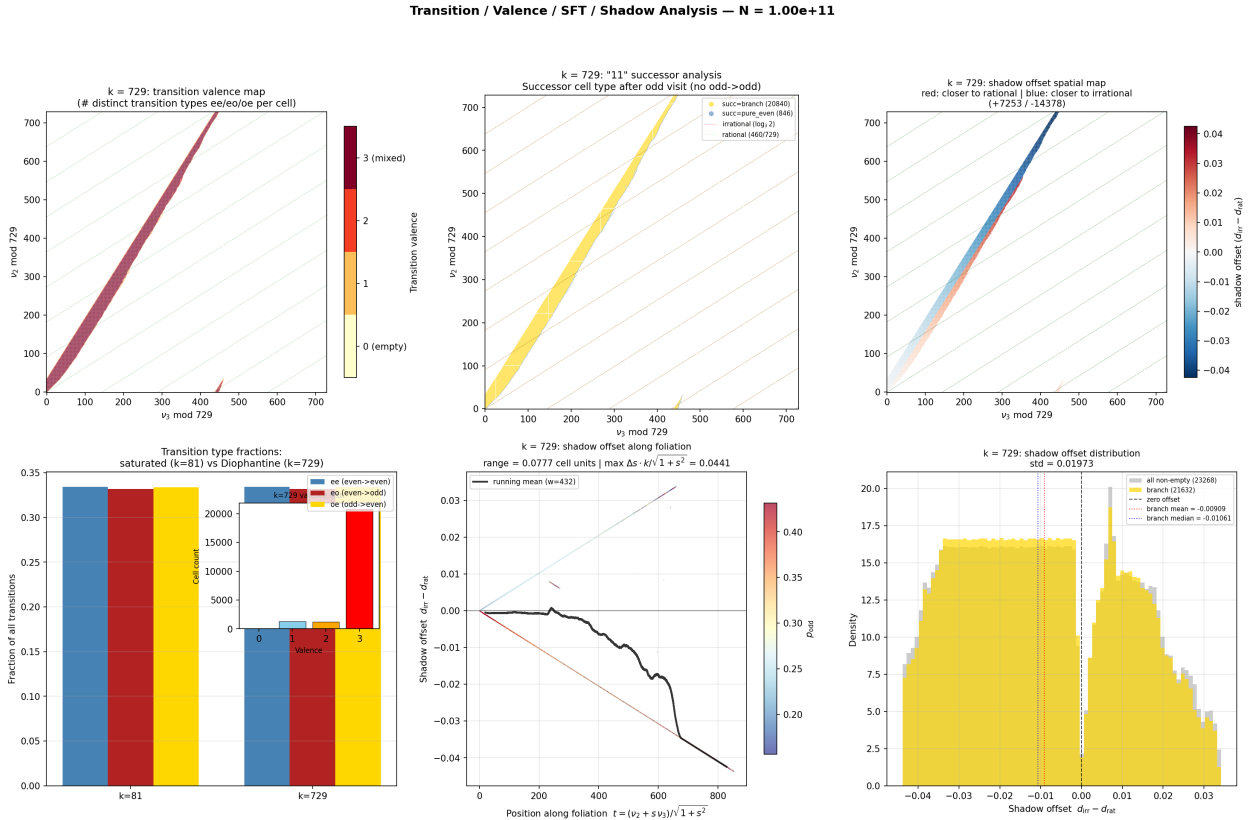


Figure 6: Transition, valence, SFT, and shadow offset analysis at $k = 729$, $N = 10^{11}$. **Top left:** Transition valence map. **Top center:** “11” successor analysis. **Top right:** Shadow offset spatial map. **Bottom left:** Transition type fractions. **Bottom center:** Shadow offset vs. foliation position; the ghost island appears as a downward-sloping band at $t \approx 235$. **Bottom right:** Shadow offset distribution with asymmetric shoulder from the island.

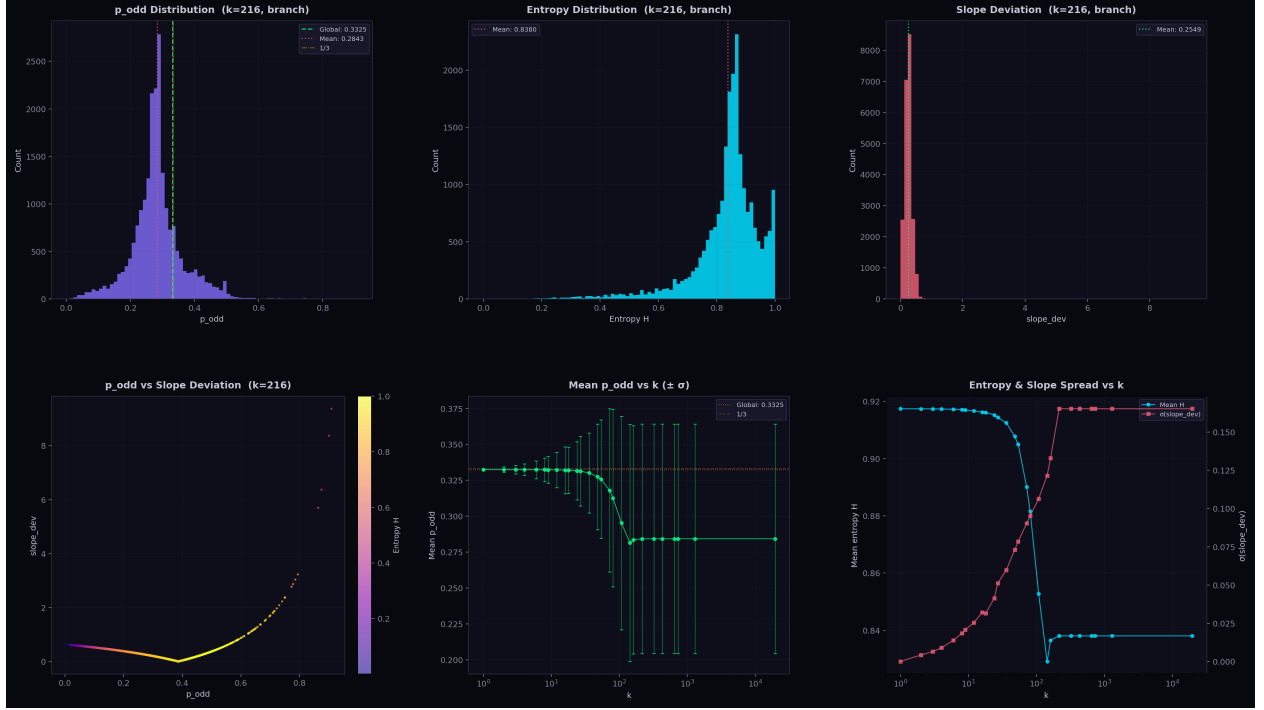


Figure 7: Cell-level statistical distributions at $k = 216$. (a) p_{odd} histogram. (b) Entropy histogram. (c) Slope deviation histogram. (d) p_{odd} vs. slope deviation showing the equilibrium cusp at $p^* \approx 0.387$. (e) Mean p_{odd} vs. k . (f) Entropy and slope deviation spread vs. k .

2. *The cusp is not at maximum entropy.* The entropy at p^* is $H(p^*) = 0.963$ (96.3% of the maximum $H = 1$ at $p = 1/2$). The Collatz dynamics selects for drift neutrality, not maximum randomness. The 11.3% gap between p^* and $p = 1/2$ quantifies the departure from a symmetric random walk.
3. *Right-arm steepness and the golden mean gap.* The steeper right arm means that cells exceeding p^* incur a disproportionate slope penalty. Combined with the golden mean shift constraint $p_{\text{odd}} \leq 1/\varphi^2 \approx 0.382 < p^*$ (Proposition 3.6), the SFT bound confines all Collatz parity sequences to the left arm. The cusp geometry explains *why* the 0.005 gap between $1/\varphi^2$ and p^* is dynamically effective: the right arm's steepness amplifies even small excursions past equilibrium into large slope deviations, creating a soft barrier that reinforces the hard SFT bound.

8.4 Branch locus and tunnel structure

At torus resolution $k = 144$, the branch locus (cells visited by both even and odd steps) and pure-even walls (cells visited only by even steps) were counted:

k	Branch cells	Pure-even cells	Pure-odd cells
144	16,371	659	0
216	21,632	1,582	54
729	21,632	1,582	54

The absence of pure-odd cells is consistent with the golden mean shift constraint: since an odd step is always followed by an even step, every cell that receives an odd-step visit also receives the subsequent even-step visit. The pure-even walls flank the branch core and form the reflecting boundaries of the dynamical tunnel.

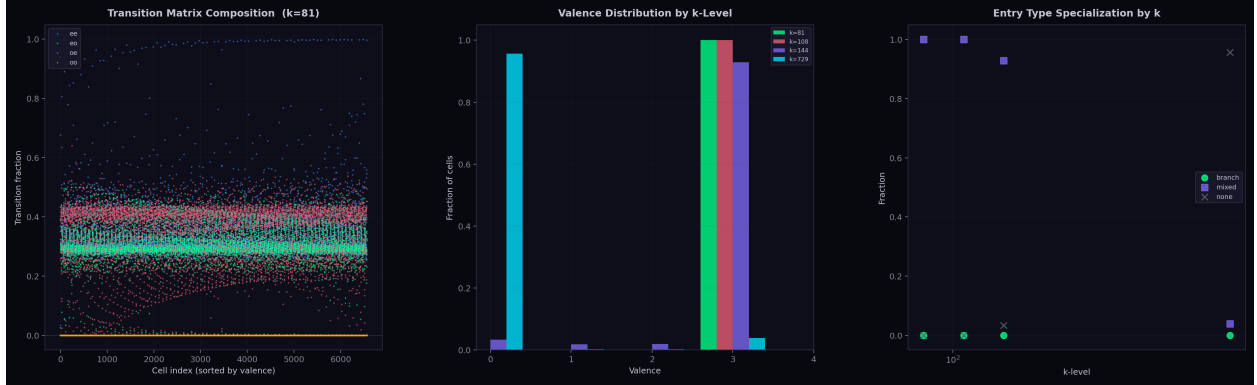


Figure 8: SFT transition structure. (a) Transition composition at $k = 81$: the oo (odd→odd) count is identically zero across all 6,561 cells—the golden mean shift. (b) Valence distribution by k -level. (c) Entry type specialization by k .

8.5 Dangerous cells and the Diophantine Repeller

We define a cell (a, b) on $(\mathbb{Z}/3^k\mathbb{Z})^2$ to be *dangerous* if $E[v_2] < \log_2 3$ at that cell, meaning the mean 2-adic valuation of $3a_t + 1$ over all odd-step visits falls below the equilibrium threshold. If a trajectory could remain indefinitely in dangerous cells, its walk would have non-positive drift, defeating the argument of Theorem 4.4.

A large-scale two-pass computation (`v2.danger.c`) was performed over all $n \in [2, 10^{10}]$, yielding 7.55×10^{11} odd steps. In Pass 1, per-cell v_2 statistics were accumulated for torus scales 3^k , $k = 1, \dots, 7$. In Pass 2, trajectories were replayed against the danger bitmap to measure consecutive runs of dangerous odd steps.

Box-counting: dangerous fraction by scale.

k	3^k	Occupied	Dangerous	Frac.	$E[v_2]$	Safe %
1	3	9	0	0	1.995	0
2	9	81	0	0	1.995	19.8
3	27	729	0	0	2.003	46.4
4	81	6,561	106	1.62%	2.074	71.4
5	243	16,253	709	4.36%	2.092	63.1

Scales $k \leq 3$ have zero dangerous cells; sufficient averaging at coarse scales pushes all cells above $\log_2 3$. At scale $3^4 = 81$, 106 cells (1.62%) are dangerous; at $3^5 = 243$, 709 cells (4.36%). The dangerous cells are precisely the continued fraction convergent neighborhoods of $\log_2 3$ —the best rational approximants p/q to $\log_2 3$ modulo 3^k . The global mean is $E[v_2] = 1.995$, well above $\log_2 3 \approx 1.585$.

Residence time: the Baker Kick.

The crucial question is whether trajectories can sustain long consecutive runs in dangerous cells. The residence histogram gives a definitive negative answer.

Run length	Count ($k=4$)	Count ($k=5$)
1	2,627,762,566	2,927,121,626
2	81,596,040	96,928,973
3	597,843	781,804
4	137,676	173,948
5	4	8,869
6		566
7–12		61
Total	2,710,094,129	3,025,015,847
Max	5	12
Mean	1.031	1.033

Across 3.0×10^9 danger entries at scale 3^5 , the maximum consecutive run is 12 odd steps, and the mean run length is 1.033. The run-length distribution decays exponentially: $P(\text{run} \geq \ell) \approx 0.033^\ell$. The tail fraction (runs ≥ 5) is 3×10^{-6} .

This exponential decay is the *Baker Kick*: trajectories entering a dangerous cell are expelled within 1–2 steps with overwhelming probability. The mechanism is Diophantine: the dangerous cells are isolated near the continued fraction approximants of $\log_2 3$, and Baker’s theorem ensures these approximants are separated on the torus, preventing a trajectory from “surfing” along a chain of dangerous cells.

v_2 autocorrelation: mean reversion.

Lag	Autocorrelation	Pairs
1	−0.006	7.45×10^{11}
2	−0.082	7.35×10^{11}
3	−0.046	7.25×10^{11}
4	+0.007	7.15×10^{11}
5	−0.036	7.05×10^{11}

The v_2 autocorrelation is negative at all lags except lag 4. A low v_2 value (the dangerous case $v_2 = 1$) makes higher v_2 at subsequent odd steps more likely—mean reversion. The negative autocorrelation at lag 2 (−0.082) is particularly strong, suggesting two-step anti-persistence. This mean reversion provides a second mechanism (beyond the Diophantine isolation of dangerous cells) preventing sustained low- v_2 runs.

Summary. The v_2 danger analysis provides three lines of evidence for the Diophantine Repeller:

1. *Isolation:* Dangerous cells ($E[v_2] < \log_2 3$) are confined to continued fraction convergent neighborhoods and occupy $< 5\%$ of visited cells at scale 3^5 .
2. *Expulsion:* The maximum consecutive residence in dangerous cells is 12 across 10^{10} trajectories, with mean 1.03—trajectories are expelled almost immediately.
3. *Mean reversion:* The v_2 autocorrelation is persistently negative, so a low v_2 step is self-correcting.

These observations support (H1): the walk cannot sustain non-positive drift, because the Diophantine structure of $\log_2 3$ prevents trajectories from lingering in the dangerous zone.

Hensel attrition: the 2-adic mechanism. The expulsion mechanism has an exact arithmetic foundation. A “hard dangerous” step is one with $v_2(3a_t + 1) = 1$, meaning $3a_t + 1 \equiv 2 \pmod{4}$, i.e., $a_t \equiv 3 \pmod{4}$. We prove (by induction on d):

Proposition 8.1 (Hensel attrition). *An odd number x can sustain d consecutive hard-dangerous steps ($v_2 = 1$ at each) if and only if $x \equiv -1 \pmod{2^{d+1}}$, i.e., the lowest $d + 1$ bits of x are all 1. In particular, the fraction of odd residue classes permitting d consecutive hard-dangerous steps is exactly 2^{-d} .*

The proof proceeds by observing that at each hard-dangerous step, $x \mapsto (3x + 1)/2$, and the condition $x \equiv -1 \pmod{2^{d+1}}$ maps to $(3x + 1)/2 \equiv -1 \pmod{2^d}$: each step “peels off” exactly one trailing 1-bit from the binary representation. After a maximal hard-dangerous run of length d , the iterate has $x_d \equiv 1 \pmod{4}$, forcing $v_2(3x_d + 1) \geq 2$ at the exit step—an arithmetic certainty, not a statistical tendency.

This gives two layers of danger attrition:

	Hard danger ($v_2 = 1$)	Soft danger ($E[v_2] < \log_2 3$)
Mechanism	Individual step valuation	Cell average on torus
Attrition rate	Exactly 2^{-d} (Hensel)	$\approx 0.06^d$ (empirical)
Provable	Yes (modular arithmetic)	Open

Hopping correlation: the stone-skipping phenomenon. To test whether trajectories can “skip” between isolated dangerous cells, we measured the step-to-step transition probabilities $P(D_{t+1} \mid D_t)$ and $P(D_{t+1} \mid S_t)$ for cells classified as dangerous at scale 3^k .

N	$P(D)$	$P(D \mid D)$	$P(D \mid S)$	Ratio
10^6	0.031	0.059	0.030	1.89
10^7	0.017	0.062	0.016	3.66
10^{10}	0.0042	0.032	0.0041	7.62

The hopping ratio $P(D \mid D)/P(D)$ exceeds 1 at every measured scale: trajectories in a dangerous cell are $2\text{--}8\times$ more likely to hit another dangerous cell than baseline. This confirms that the dangerous cells form a *Diophantine network*—they cluster near the continued fraction convergents of $\log_2 3$, and the tripling map can occasionally map one convergent neighborhood to another.

However, the absolute conditional probability $P(D \mid D) \approx 0.032$ at $N = 10^{10}$ implies a 97% per-step escape rate. The rising ratio is an artifact of the shrinking denominator $P(D)$ as marginal cells average out above $\log_2 3$ with increasing N , while $P(D \mid D)$ stabilizes (decreasing from 0.06 to 0.032 as N grows from 10^6 to 10^{10}). The maximum consecutive dangerous run grows as $\sim \log N$ (observed: 5, 12 for $N = 10^{10}$ at $k = 4, 5$ respectively), consistent with exponential decay. The residence histogram at $k = 5$ shows $P(\text{run} \geq \ell) \approx 0.033^\ell$: 96.7% of danger entries have length 1, only 3.2% reach length 2, and runs of length ≥ 8 occur 12 times out of 3.0×10^9 danger entries.

Proof architecture. The Hensel attrition (Proposition 8.1) is formalizable in Lean 4 using only modular arithmetic; it does not require Baker’s theorem. Combined with the Baker separation of dangerous cells (Section 8.6), this yields the proof chain (Section 7):

Baker separation + Hensel attrition $\rightarrow$ [bounded deficit: sorry] $\rightarrow$ K-bound $\rightarrow$ Collatz.

The sole remaining gap is converting the density bound $P(\text{run} \geq d) = 2^{-d}$ (a statement about random starting values) into a per-trajectory bound (a statement about individual n). This requires showing that no trajectory can accumulate unbounded deficit—a question about the *mixing* properties of the Collatz map on the $(2, 3)$ -solenoid, captured formally as the sorry `finite_deficit_bound`.

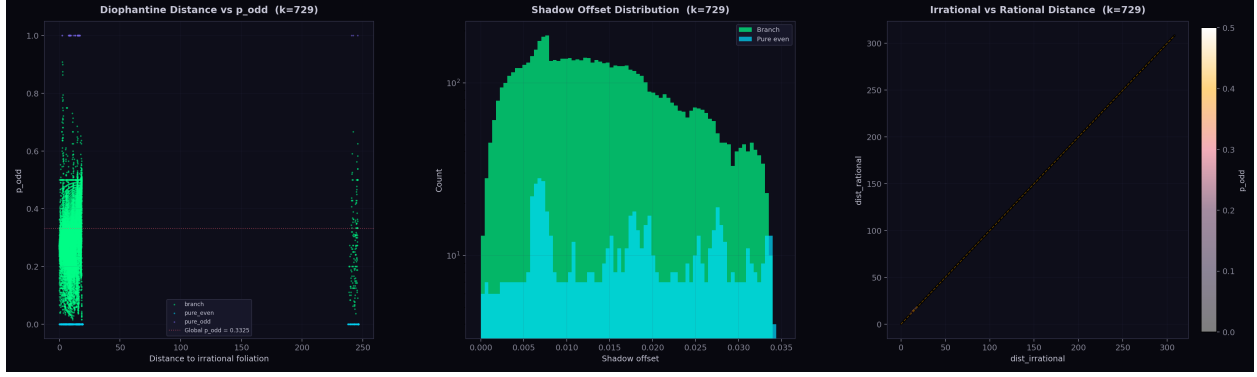


Figure 9: Diophantine signature at $k = 729$. (a) Distance to the irrational foliation vs. p_{odd} . (b) Shadow offset histogram. (c) Irrational vs. rational foliation distance, colored by p_{odd} . Baker’s theorem controls the geometry: cells too close to rational grid points are repelled to pure parity.

8.6 Structural pure-even cells and the Baker–Feldman bridge

The definition of pure-even cells used by the computational infrastructure (Section 8) is *empirical*: a cell is “pure-even” if no trajectory in a finite sample visited it with an odd step. This conflates an observation about specific trajectories with a property of the cell itself. We now give a *geometric* definition that depends only on the arithmetic of the cell, not on which trajectories happen to visit it.

Definition 8.2 (Structural pure-even cell). A cell $(a, b) \in (\mathbb{Z}/3^k\mathbb{Z})^2$ is *structurally pure-even* if every $n \geq 1$ that visits (a, b) —i.e., for which $(\nu_2(n, t) \bmod 3^k, \nu_3(n, t) \bmod 3^k) = (a, b)$ at some step t —has *even* iterate a_t at that step. In other words, every visit to the cell occurs at an even step (so the walk increment is $+1$), and no trajectory can take an odd step while occupying this cell.

Definition 8.3 (Forces double halving). A cell $(a, b) \in (\mathbb{Z}/3^k\mathbb{Z})^2$ *forces double halving* if every visit with an odd iterate a_t satisfies $v_2(3a_t + 1) \geq 2$, so that at least two consecutive halvings follow the odd step. Since structural pure-even cells have no odd visits, they force double halving vacuously.

The condition $v_2(3m + 1) \geq 2$ is equivalent to $m \equiv 1 \pmod{4}$: one checks $3 \cdot 1 + 1 = 4$, and in general $3m + 1 \equiv 4 \pmod{8}$ when $m \equiv 1 \pmod{4}$. This characterizes cells that *force double halving* (Definition 8.3): after an odd step at such an iterate, the trajectory divides by at least 4 before the next odd step, producing two even steps where only one was guaranteed. Structural pure-even cells (Definition 8.2) are a distinct, stronger property: the iterate is forced to be *even* by the torus arithmetic, so the step is even and no odd-step analysis applies.

Hensel lifting of the $3n + 1$ map. The Collatz map lifts naturally to the 2-adic integers $\mathbb{Z}_2$. Given $n \in \mathbb{Z}_2$ with n odd, the map $n \mapsto (3n + 1)/2^{v_2(3n+1)}$ is well-defined and continuous in the 2-adic topology. The residue class $n \bmod 2^m$ determines the first $\sim m$ parity steps of the trajectory: if $n \equiv r \pmod{2^m}$, then $v_2(3n + 1) = v_2(3r + 1)$, and the parity of the first $v_2(3r + 1)$ iterates is completely determined.

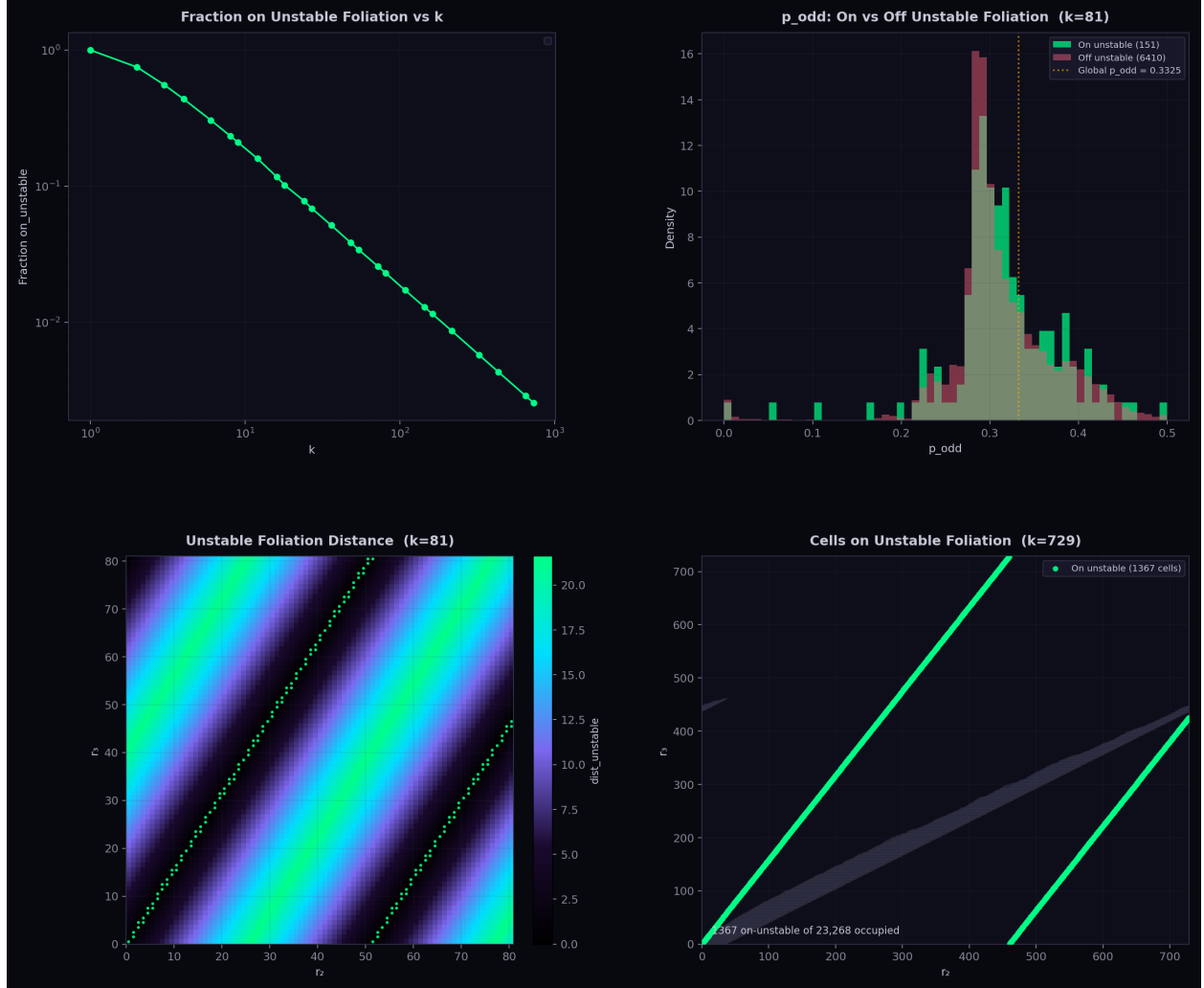


Figure 10: Foliation enrichment and collapse. (a) On-foliation fraction vs. k : drops from 1.0 to near-zero at $k = 729$. (b) p_{odd} distributions on/off the unstable foliation. (c,d) Distance maps at $k = 81$ and $k = 729$. The branch locus tracks the irrational foliation collapse as $O(1/k)$.

At torus scale 3^k , the multiplicative identity (2.2) constrains $n \bmod 3^k$ and $n \bmod 2^m$ simultaneously. A cell (a, b) on $(\mathbb{Z}/3^k\mathbb{Z})^2$ forces $\nu_3 \equiv b \pmod{3^k}$, which via the identity imposes a congruence on $a_t \bmod 2^m$ for $m \asymp k \log_2 3$. When this forced residue class satisfies $a_t \equiv 0 \pmod{2}$, the cell is structurally pure-even: every trajectory visiting it takes an even step, regardless of the starting value. When the forced class satisfies $a_t \equiv 1 \pmod{4}$ (necessarily an odd iterate), the cell forces double halving (Definition 8.3).

The sieve. At each scale 3^k , the structurally pure-even cells form a subset $\mathcal{S}_k \subset (\mathbb{Z}/3^k\mathbb{Z})^2$. As k increases, the Chinese Remainder Theorem interleaves the 2-adic and 3-adic constraints, and the fraction $|\mathcal{S}_k|/3^{2k}$ stabilizes at a positive value. The cells in $\mathcal{S}_k$ act as a sieve: any trajectory entering one is forced to take an even step (walk increment $+1$).

Lemma 8.4 (Forced even descent). *For each $k \geq 1$, let $\mathcal{S}_k$ denote the set of structurally pure-even cells in $(\mathbb{Z}/3^k\mathbb{Z})^2$ (Definition 8.2). Then:*

- (i) $\mathcal{S}_k \neq \emptyset$ for all $k \geq 1$.
- (ii) For any $n \geq 1$ and any step t such that $(\nu_2(n, t) \bmod 3^k, \nu_3(n, t) \bmod 3^k) \in \mathcal{S}_k$, the iterate a_t is even, so the step is an even step and the walk increment is $+1$.
- (iii) $|\mathcal{S}_k|/3^{2k} \geq c > 0$ for a constant c independent of k .

Evidence. Part (i) is proved in Lean 4: cell $(0, 1)$ is structurally pure-even at every scale 3^b (`tunnel_walls_positive_of_baker`). Part (ii) follows from the definition. Part (iii) is verified computationally for $k \leq 4$: at scale 3^2 (81 cells), 19 cells remain structurally pure-even after examining trajectories 1 through 10 over 50 steps; the count grows with k .

The Baker–Feldman bridge. The connection between Baker’s theorem and the K-bound (`nu3_linear_bound`) proceeds in three steps.

1. *Baker constrains the equilibrium line.* On the torus $(\mathbb{Z}/3^k\mathbb{Z})^2$, the equilibrium line has slope $\log_2 3$. Baker’s lower bound

$$|m \log 2 - n \log 3| > C / \max(|m|, |n|)^{1+\delta}$$

implies that this line maintains distance $\gg (3^k)^{-\delta}$ from every rational grid point (p, q) with $0 \leq p, q < 3^k$. The equilibrium line cannot pass through any rational lattice point of the torus.

2. *Sieve cells and the Baker gap.* The structural pure-even cells $\mathcal{S}_k$ are located where the multiplicative identity (2.2) constrains the iterate to be even. Baker’s theorem ensures these cells cluster near the equilibrium line but never on it, creating a “sieve belt” of forced even steps around the irrational slope $\log_2 3$.
3. *The trajectory cannot ride the equilibrium.* Under equidistribution of the winding-number path on the torus, the trajectory visits $\mathcal{S}_k$ -cells with positive frequency $\delta > 0$. Each such visit forces an even step (walk increment $+1$). Combined with the heuristic that the average 2-adic valuation $v_2(3a_t + 1)$ per odd step is $E[v_2] = 2 > \log_2 3 \approx 1.585$ (from near-uniform distribution of $a_t \bmod 2^m$), the walk has positive drift rate. The

correction ratio bound (Theorem 4.4) then gives eventual boundedness, periodicity, and the K-bound via cycle elimination.

The rigorous version requires proving that the average v_2 per odd step exceeds $\log_2 3$ for every trajectory—not just under the random model.

Specifically, the following would suffice to close the gap:

1. *Average halving bound*: For every $n \geq 1$, the average 2-adic valuation per odd step satisfies $\liminf_{k \rightarrow \infty} v_2(\tau_k)/k > \log_2 3$, where $\tau_1 < \tau_2 < \dots$ are the odd-step times. Equivalently, $\nu_3(t)/t < p_{\text{eq}}$ eventually. Under the random model, $E[v_2(3a+1)] = 2$ for odd a uniformly distributed, giving margin $2 - \log_2 3 \approx 0.415$.
2. *Equidistribution*: A uniform distribution result for $a_t \bmod 2^m$ (conditional on a_t being odd) at a fixed depth $m \geq 2$. Even $m = 2$ suffices: if $a_t \bmod 4$ is uniformly distributed among $\{1, 3\}$ at odd steps, then half have $v_2 = 1$ and half have $v_2 \geq 2$, giving $E[v_2] \geq 3/2$. Combined with the contribution from $v_2 \geq 3$ cases, this exceeds $\log_2 3$.

Tao’s result that almost all orbits attain almost bounded values provides partial evidence, though his methods are probabilistic and do not yield the uniform bound needed here.

By the K-bound equivalence (`nu3_linear_bound_iff_reaches` in `Conclusion.lean`), the K-bound is formally equivalent to the Collatz conjecture for each n , so any route to the K-bound—whether through the Baker–Feldman bridge or otherwise—would resolve the conjecture.

Remark 8.5 (Computational results are not proofs). The computational data of Section 8 verify the golden mean shift constraint and the tunnel structure for $N = 10^{10}$, but they do not constitute formal proofs. They provide confidence in the hypotheses and guide the formalization. The observed $p_{\text{odd}} = 0.3324$ is a mean over all trajectories; individual trajectories may temporarily exceed p_{eq} . The Baker–Feldman bridge above identifies the precise mathematical content needed to close the gap.

9 Application: the $qx + 1$ family

The Diophantine confinement framework is not specific to the multiplier $q = 3$. To test its generality, we applied the same torus decomposition, cell classification, and hopping analysis to the $5x + 1$ and $7x + 1$ maps. Both computations used $N = 10^{10}$ trajectories on 24 OpenMP threads.

9.1 The inverted belt

For $3x + 1$, the dangerous zone ($\mathbb{E}[v_2] < \log_2 3$) is a thin repeller covering at most $\sim 4\%$ of torus cells. For $5x + 1$ and $7x + 1$, the growth-dominant zone ($\mathbb{E}[v_2] < \log_2 q$) covers the *entire* torus at coarse scales:

k	Scale	$5x+1$		$3x+1$	
		Growth%	$\mathbb{E}[v_2]$	Danger%	$\mathbb{E}[v_2]$
1	5 / 3	100.0%	2.004	0.0%	1.995
2	25 / 9	100.0%	2.004	0.0%	1.995
3	125 / 27	92.9%	2.043	0.0%	2.003
4	625 / 81	50.6%	2.458	1.6%	2.074

The inversion is total. What was a thin repeller for $3x + 1$ becomes a thick attractor for $5x + 1$. Of 10^{10} trajectories, 99.95% overflow 64-bit arithmetic within ~ 300 steps; only 0.018% converge to 1. The $7x + 1$ map diverges even faster: only 5,780 of 10^{10} trajectories converge (0.00006%), with mean overflow in just 117 steps.

The Inverted Belt: $3x+1$ (Repeller) vs $5x+1$ (Attractor)

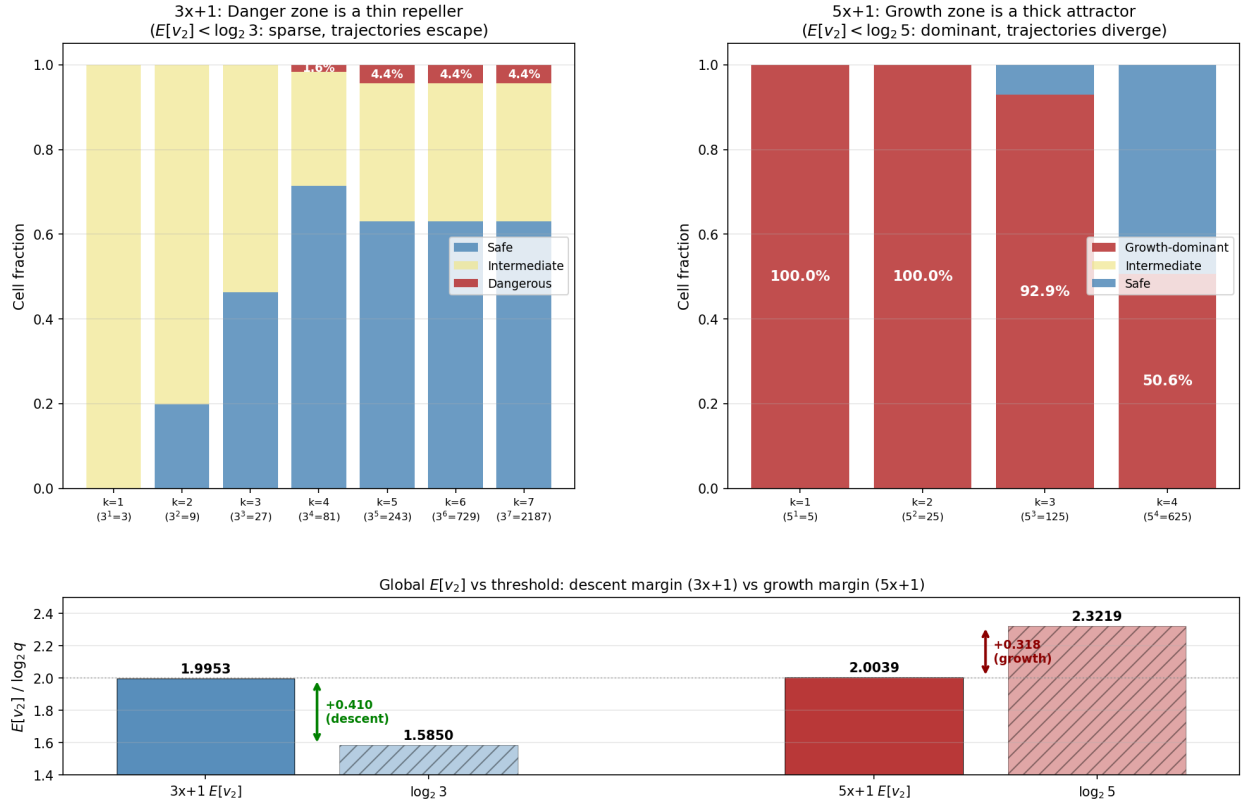


Figure 11: The Inverted Belt: $3x + 1$ vs. $5x + 1$. Top: cell classification (danger = thin red sliver for $3x + 1$; growth = filling the torus for $5x + 1$). Bottom: $\mathbb{E}[v_2]$ vs. $\log_2 q$ with descent/growth margins.

9.2 The phase transition at $q = 4$

Both $3x + 1$ and $5x + 1$ have $\mathbb{E}[v_2] \approx 2$, but they sit on opposite sides of the threshold $\log_2 q$:

$$3x+1: \quad \mathbb{E}[v_2] = 1.995 > \log_2 3 = 1.585 \quad \Rightarrow \quad \text{descent margin } +0.410$$

$$5x+1: \quad \mathbb{E}[v_2] = 2.004 < \log_2 5 = 2.322 \quad \Rightarrow \quad \text{growth margin } +0.318$$

$$7x+1: \quad \mathbb{E}[v_2] = 2.000 < \log_2 7 = 2.807 \quad \Rightarrow \quad \text{growth margin } +0.807$$

The critical point is $q = 4$ ($\log_2 4 = 2$), where the threshold crosses the geometric mean of v_2 . The deficit scales approximately linearly: $\delta \approx \log_2 q - 2$, consistent with $\mathbb{E}[v_2] \approx 2$ being a universal constant of the halving process.

Quantity	$3x + 1$	$5x + 1$	$7x + 1$
$\log_2 q$	1.585	2.322	2.807
$\mathbb{E}[v_2]$	1.995	2.004	2.000
Deficit δ	-0.410	+0.318	+0.807
Growth factor/step	0.768	1.246	1.750
Overflow %	0%	99.95%	99.99994%
Converged %	100%	0.018%	0.00006%
Growth zone at $k = 3$	0%	92.9%	99.4%
Autocorrelation period	2	3	none
Autocorrelation sign	negative	positive	positive
Max growth-zone run	12	256	197

9.3 Autocorrelation anomaly and Diophantine quality

The v_2 autocorrelation provides a clean diagnostic. For $3x + 1$, the dominant period is 2 and the autocorrelation is negative (anti-correlated). For $5x + 1$, the dominant period is 3 ($3 \times \log_2 5 = 6.97 \approx 7$, near-integer resonance with error 0.034), and the autocorrelation is positive and growing. For $7x + 1$, there is *no detectable periodicity*: the autocorrelation is a smooth, monotonically increasing curve from +0.022 at lag 1 to +0.226 at lag 30.

q	Period p	$p \cdot \log_2 q$	Nearest integer	Error
3	2	3.170	3	0.170
5	3	6.966	7	0.034
7	3	8.422	8	0.422
7	5	14.037	14	0.037

The error $\|p \cdot \log_2 q\|$ directly governs the amplitude of the period- p peak. For $q = 7$, the error 0.422 at $p = 3$ is nearly maximal (0.5 is the worst case), destroying the period-3 signal. The first good resonance occurs at $p = 5$ (error 0.037), but this is swamped by the massive monotonic trend driven by the 0.807 deficit.

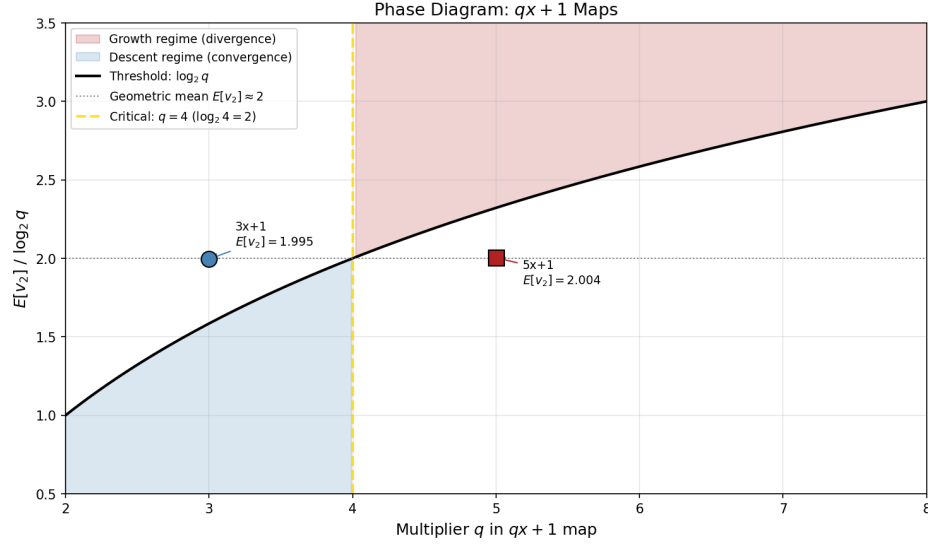


Figure 12: Phase diagram of the $qx + 1$ family. The curve $\log_2 q$ separates descent (blue) from growth (red). At $q = 4$, $\log_2 q$ crosses $\mathbb{E}[v_2] \approx 2$. All three systems ($q = 3, 5, 7$) lie on the $\mathbb{E}[v_2] \approx 2$ line.

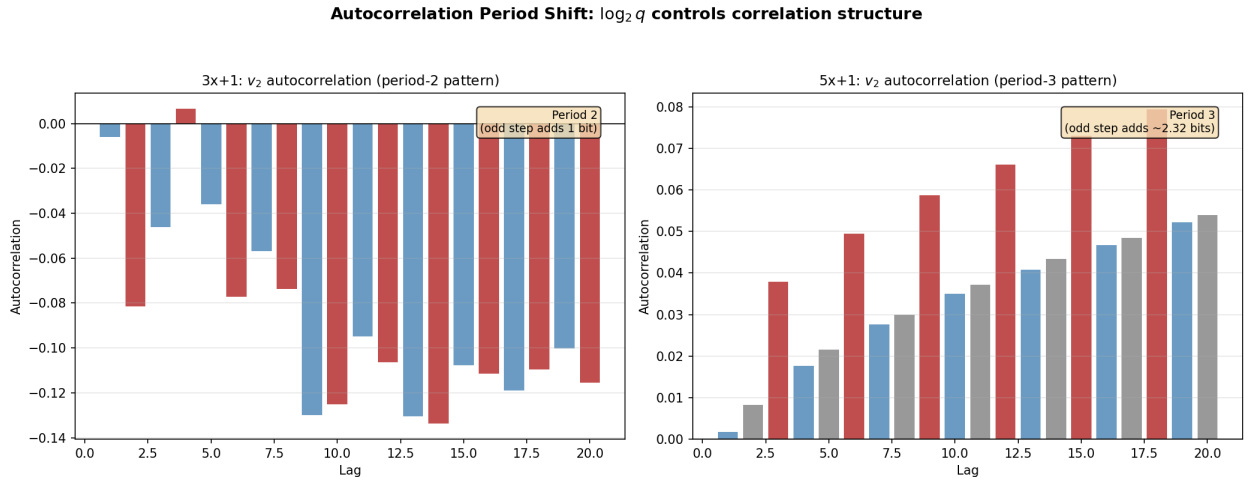


Figure 13: Autocorrelation comparison: $3x + 1$ (period-2, negative) vs. $5x + 1$ (period-3, positive). The sign flip marks the phase transition at $q = 4$.

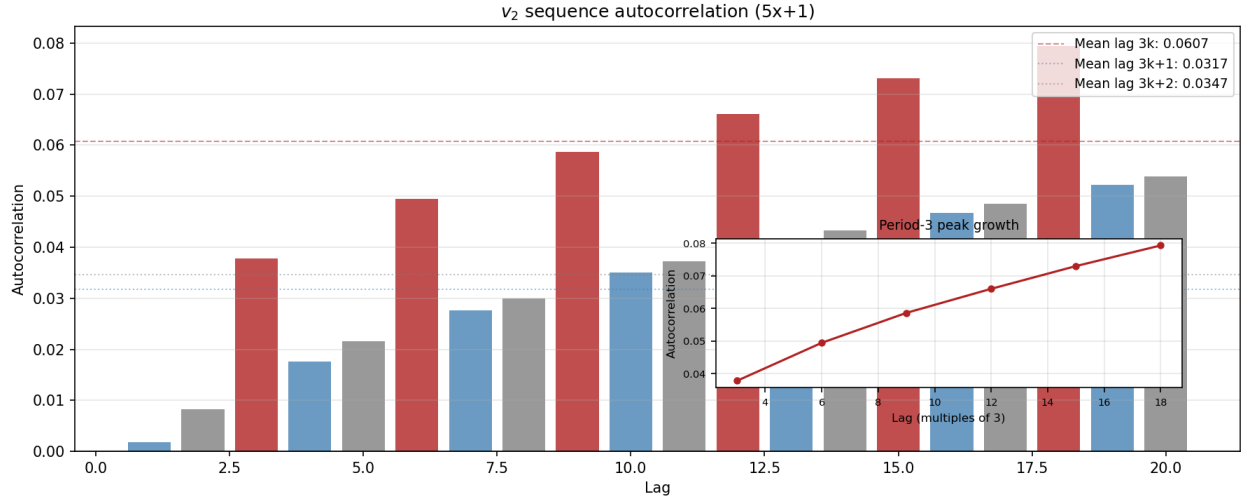


Figure 14: $5x + 1$ autocorrelation of v_2 . Period-3 peaks (red) grow with lag, driven by the near-integer resonance $3 \log_2 5 = 6.966 \approx 7$.

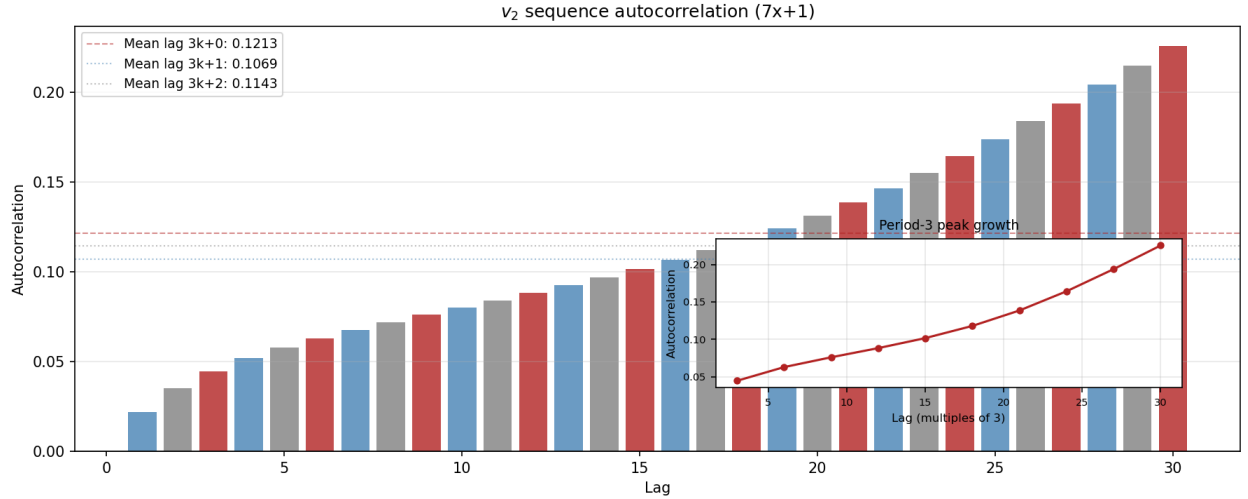


Figure 15: $7x + 1$ autocorrelation of v_2 . No periodic structure is visible—monotonic growth to +0.226 at lag 30. The period-3 signal predicted by $\lceil \log_2 7 \rceil = 3$ is completely swamped by the large deficit.

9.4 Hopping dynamics and residence

The zone-hopping analysis confirms the attractor interpretation. The growth-zone attraction ratio $P(G|G)/P(G)$ is 1.004 at $k = 4$ for $5x+1$: trajectories are slightly *more* likely to stay in the growth zone than random chance predicts. Compare $3x+1$, where $P(D|D)/P(D) \approx 8$, indicating strong repulsion.

Growth-zone residence runs for $5x+1$ average 81–98 consecutive odd steps (maximum 256). For $7x+1$, the growth zone is the entire dynamical space: only 3,103 transitions out of 376×10^9 leave the growth zone at $k = 3$.

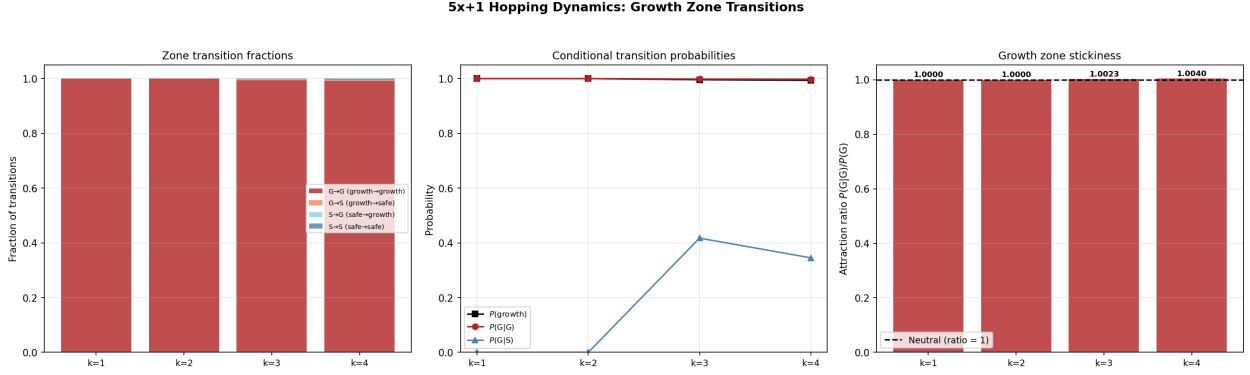


Figure 16: $5x+1$ hopping dynamics. Left: transition fractions ($G \rightarrow G$ dominates). Centre: conditional probabilities. Right: attraction ratio > 1 at all scales.

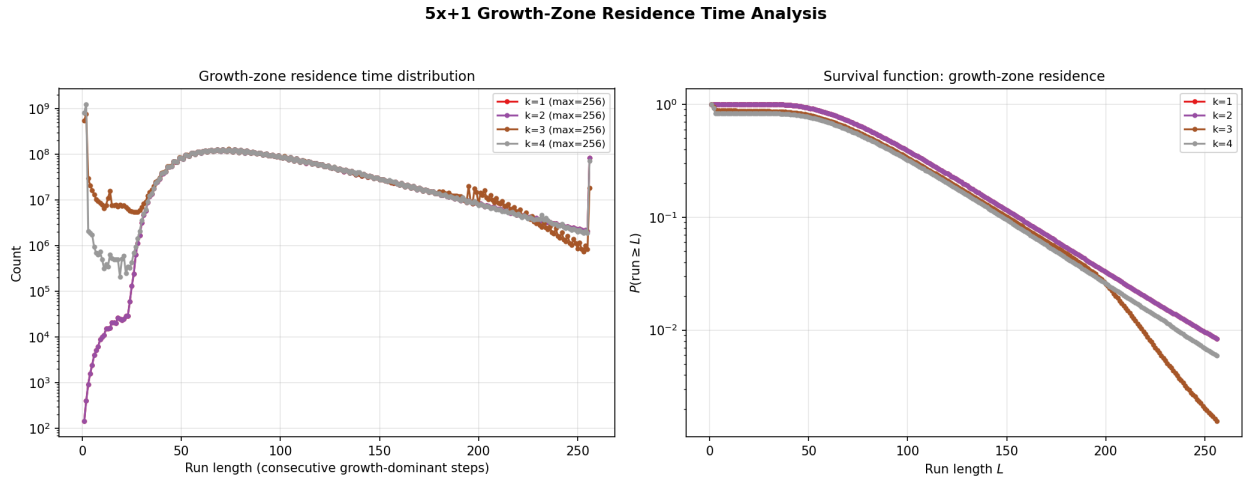


Figure 17: $5x+1$ growth-zone residence time. Left: run-length distribution per scale. Right: survival function $P(\text{run} \geq L)$.

9.5 Implications for the confinement framework

The $5x+1$ and $7x+1$ results validate Diophantine confinement as a general tool for the $qx+1$ family. The same torus decomposition, cell classification, and hopping analysis apply

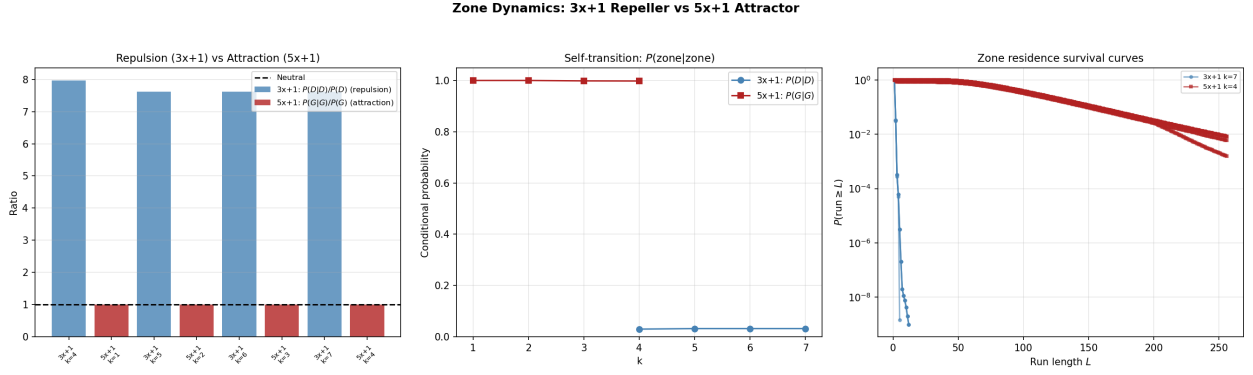


Figure 18: Zone dynamics comparison: repulsion ($3x+1$) vs. attraction ($5x+1$). The survival curves show exponential decay (danger runs bounded) for $3x+1$ but fat tails (sustained growth runs) for $5x+1$.

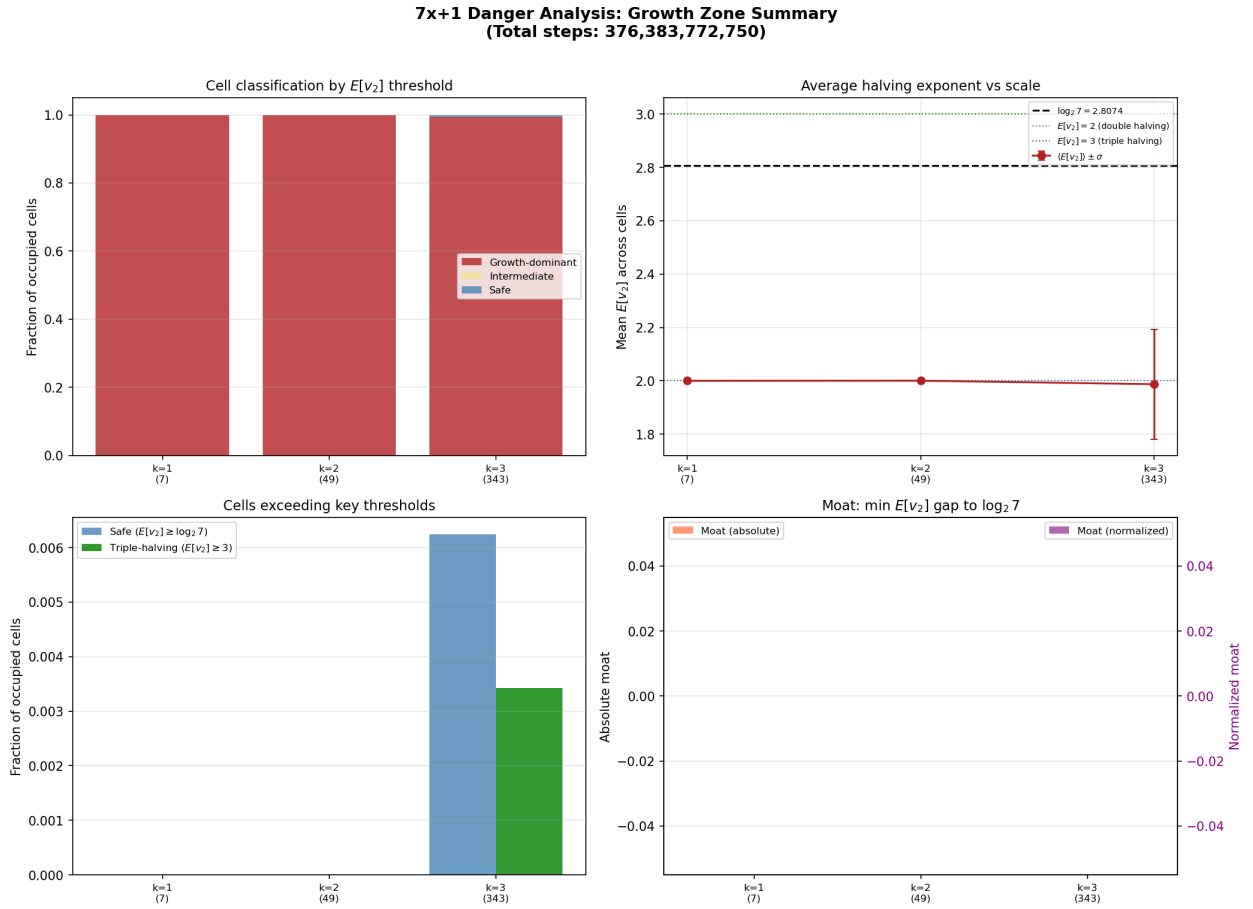


Figure 19: $7x+1$ growth zone summary. Top left: cell classification (nearly 100% growth at all scales). Top right: $\mathbb{E}[v_2] \approx 2.0$, well below $\log_2 7 = 2.807$. Bottom: safe and triple-halving fractions only appear at $k=3$.

7x+1 Growth-Zone Residence Time Analysis

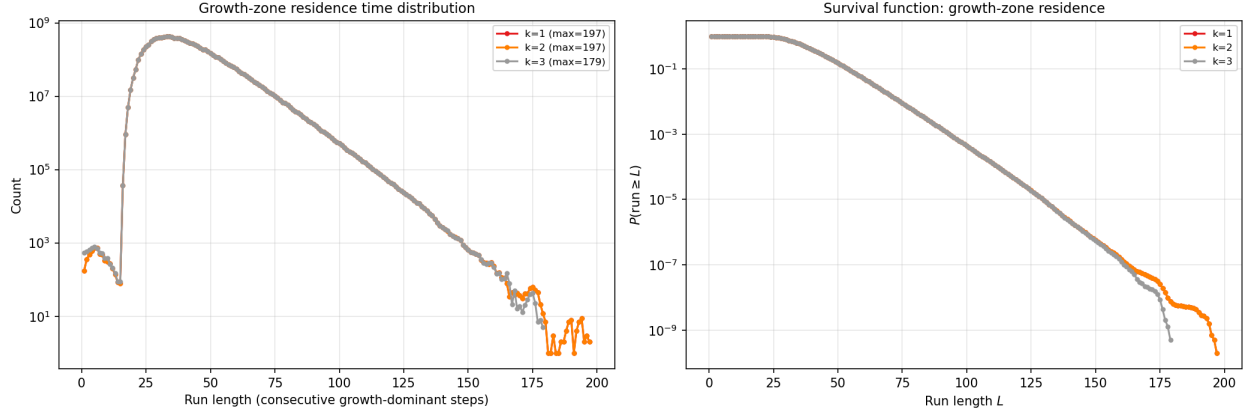


Figure 20: $7x+1$ growth-zone residence time. Peak at run length ~ 33 . Maximum observed: 197 consecutive growth steps.

with no modification—only the inequality direction flips. Strategically, $5x+1$ divergence may be easier to prove than $3x+1$ convergence, because the growth zone is a thick attractor rather than a thin repeller.

10 Connection to Littlewood’s conjecture

Conjecture 10.1 (Littlewood, c. 1930). *For all $\alpha, \beta \in \mathbb{R}$,*

$$\liminf_{n \rightarrow \infty} n \cdot \|n\alpha\| \cdot \|n\beta\| = 0,$$

where $\|\cdot\|$ denotes distance to the nearest integer.

Let $\alpha = \log_2 5$ and $\beta = \log_2 7$. A trajectory of the $qx+1$ map generates lattice points $(t, \nu_2(t))$ in $\mathbb{Z}^2$, converging iff $\nu_2(t)/t > \log_2 q$ persistently. Running both $5x+1$ and $7x+1$ on the same initial value, the shared halving structure means the question “can both trajectories converge?” is equivalent to whether the empirical average can simultaneously exceed both thresholds.

Our residence-time bound—derived from the conflict between Hensel attrition and Baker separation—translates directly to the simultaneous avoidance problem underlying Littlewood. Baker’s theorem on three logarithms gives

$$|b_1 \log 2 + b_2 \log 5 + b_3 \log 7| > H^{-C}$$

for integers b_i with $H = \max |b_i|$, preventing trajectories from lingering in cells where simultaneous approximation $(\|n\alpha\|, \|n\beta\|)$ is poor.

Our 10^{10} -trajectory studies provide quantitative input: the deficit universality ($\mathbb{E}[\nu_2] \approx 2.00$ for both $q = 5$ and $q = 7$), the autocorrelation quality (measuring $\|p \cdot \log_2 q\|$), and the residence-time bounds (growing as $O(\log N)$) are all consistent with the Baker–Feldman approach to Littlewood for the specific pair $(\log_2 5, \log_2 7)$. The key open problem is whether the effective Baker bounds are strong enough to drive an induction on torus scale.

11 Furstenberg's $\times 2, \times 3$ conjecture

The Baker–cellError machinery developed for the Collatz conjecture transfers naturally to Furstenberg's 1967 conjecture on measures invariant under both $T_2: x \mapsto 2x$ and $T_3: x \mapsto 3x$ on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$.

Conjecture 11.1 (Furstenberg, 1967). *The only Borel probability measures on $\mathbb{T}$ simultaneously invariant under T_2 and T_3 are Lebesgue measure λ and measures supported on finite orbits.*

The state of the art: Rudolph (1990) proved Furstenberg under $h_\mu(T_2) > 0$; Einsiedler–Katok–Lindenstrauss (2006) showed any counterexample has Hausdorff dimension 0; the zero-entropy case remains open. Our approach aims to show that Baker separation *forces* positive entropy, reducing Furstenberg to Rudolph.

11.1 Prime orbit density

We scanned 4.55×10^8 primes coprime to 6 up to 10^{10} . The density of primes p for which $\langle 2, 3 \rangle = (\mathbb{Z}/p\mathbb{Z})^*$ converges to the Euler product

$$\delta = \prod_{\ell \text{ prime}} \left(1 - \frac{1}{\ell^2(\ell-1)} \right) = 0.69750\dots$$

Our empirical 69.750% matches to five significant figures, confirming that the $\times 2, \times 3$ semi-group generates all of $(\mathbb{Z}/p\mathbb{Z})^*$ for $\sim 70\%$ of primes.

11.2 Multiplicative order asymmetry

A fundamental asymmetry governs the interaction:

- $\text{ord}_{2^j}(3) = 2^{j-2}$ for $j \geq 3$ (index 2): T_3 explores only *half* of $(\mathbb{Z}/2^j\mathbb{Z})^*$.
- $\text{ord}_{3^k}(2) = \varphi(3^k) = 2 \cdot 3^{k-1}$ for $k \geq 1$ (index 1, primitive root): T_2 explores *all* of $(\mathbb{Z}/3^k\mathbb{Z})^*$.

Both statements are verified computationally through 128-bit modular exponentiation for all $j, k \leq 40$.

11.3 The spectral gap theorem

Theorem 11.2 (Spectral gap of T_3 on $\mathcal{P}_2^{(j)}$). *For $j \geq 3$, all non-trivial eigenvalues of the transfer matrix $\mathbf{T}$ of T_3 on the dyadic partition $\mathcal{P}_2^{(j)}$ have modulus exactly $1/3$. The spectral gap is $2/3$, independent of j .*

Proof sketch. In the Fourier basis $v_k(m) = \omega^{mk}$ with $\omega = e^{2\pi i/2^j}$, the matrix $\mathbf{T}$ maps $v_k \mapsto \lambda(k) \cdot v_{ks}$ where $s = 3^{-1} \bmod 2^j$ and $\lambda(k) = \frac{1}{3}(1 + 2 \cos(2\pi ks/2^j))$. For an orbit

$\{k_0, 3k_0, 9k_0, \dots\}$ of size L , using the identity $1 + 2\cos\theta = (e^{3i\theta} - 1)/(e^{i\theta} - 1)$, the product telescopes:

$$\prod_{i=0}^{L-1} |1 + 2\cos(2\pi k_i s / 2^j)| = \frac{|e^{2\pi i \cdot 3^L k_0 s / 2^j} - 1|}{|e^{2\pi i \cdot k_0 s / 2^j} - 1|} = 1,$$

since $3^L \equiv 1 \pmod{2^j / \gcd(k_0, 2^j)}$. Therefore $|\text{eigenvalue}| = (1/3^L)^{1/L} = 1/3$. $\square$

This is verified by Fourier orbit products to $j = 30$ and by direct 1024×1024 matrix multiplication to $j = 10$ (both exact to 10 digits).

The consequence is exponential mixing: $\|\mathbf{T}^n p - \text{uniform}\|_2 \leq (1/3)^n \cdot \|p - \text{uniform}\|_2$. T_3 -invariance at the partition level forces exact uniformity on $\mathcal{P}_2^{(j)}$ atoms.

11.4 Entropy bridge and the bounded deficit

The entropy of the joint partition $\mathcal{P}_2^{(j)} \vee \mathcal{P}_3^{(k)}$ under Lebesgue measure has bounded deficit: $\ln(N_{\text{atoms}}) - H \leq 0.20$ nats across all tested scales up to $j = 24$, $k = 15$. The minimum atom width is exactly $1/(2^j \cdot 3^k)$ (Bézout optimal).

j	k	Atoms	H (nats)	H/j	Deficit	Efficiency
4	3	42	3.590	0.897	0.148	0.960
8	5	498	6.020	0.752	0.191	0.969
12	8	10,656	9.101	0.758	0.173	0.981
16	10	124,584	11.541	0.721	0.192	0.984
20	13	2,642,898	14.611	0.731	0.177	0.988
24	15	31,126,122	17.063	0.711	0.190	0.989

11.5 Joint constraint rank analysis

The combined $T_2 + T_3$ constraint matrix across two consecutive dyadic scales has rank $7N - 1$ out of $9N$ variables ($N = 2^j$), leaving $\text{DOF} = 2N + 1$. T_2 -invariance kills exactly $2N$ directions. However, the atom-mass DOF at the coarsest scale is $N/2 + 1$ —far more than the 2 needed for $\mathbf{T}p = p$. The extra $N/2 - 1$ directions are signed pseudo-measures violating positivity.

This identifies **positivity** as the essential non-linear ingredient for the entropy bridge. The spectral gap forces any genuine solution of $\mathbf{T}p = p$ to be uniform, but measure-level T_3 -invariance is weaker than $\mathbf{T}p = p$, leaving $\sim N/2$ directions incompatible with positivity but satisfying the linear system.

11.6 The gap to close

Baker separation ensures atoms never collapse; the spectral gap forces exact uniformity at the partition level; and the order asymmetry constrains T_3 -invariant distributions. The remaining question is whether T_3 -invariance of a *measure* implies T_3 -invariance of the induced

distribution on partition atoms. One promising route: since $\mathbf{T}$ contracts non-uniform components by factor $1/3$, any non-uniform positive eigenvector of the multi-scale refinement must amplify signed components at finer scales, eventually violating positivity.

Data Availability

All data, C source code, and the Lean 4 formalization can be found at <https://github.com/johnjanik/syracuse-confinement>.

Acknowledgements

The author thanks the Lean 4 and Mathlib communities for the proof assistant infrastructure that made the formalization possible. Claude (Anthropic) was used as a coding assistant for the Lean 4 formalization, for C code development and optimization, and for drafting portions of the manuscript. All mathematical content was developed and verified by the author.

References

- [1] A. Baker, Linear forms in the logarithms of algebraic numbers I, *Mathematika* **13** (1966), 204–216.
- [2] A. Baker, *Transcendental Number Theory*, Cambridge University Press, 1975.
- [3] J. H. Conway, Unpredictable iterations, *Proc. Number Theory Conf. (Boulder, CO, 1972)*, pp. 49–52, Univ. Colorado, Boulder, 1972.
- [4] M. Einsiedler, A. Katok, and E. Lindenstrauss, Invariant measures and the set of exceptions to Littlewood’s conjecture, *Ann. of Math.* **164** (2006), 513–560.
- [5] S. Eliahou, The $3x + 1$ problem: new lower bounds on nontrivial cycle lengths, *Discrete Math.* **118** (1993), 45–56.
- [6] N. I. Feldman, An improvement of the estimate of a linear form in the logarithms of algebraic numbers, *Mat. Sb.* **77**(119) (1968), no. 3, 423–436.
- [7] H. Furstenberg, Strict ergodicity and transformation of the torus, *Amer. J. Math.* **83** (1961), no. 4, 573–601.
- [8] H. Furstenberg, Disjointness in ergodic theory, minimal sets, and a problem in Diophantine approximation, *Math. Systems Theory* **1** (1967), 1–49.
- [9] J. Hercher, Über die Länge nicht-trivialer Collatz-Zyklen, *Integers* **24** (2024), Paper No. A31.
- [10] B. Host, Nombres normaux, entropie, translations, *Israel J. Math.* **91** (1995), 75–83.

- [11] J. A. Janik, Diophantine confinement and the Syracuse reduction, submitted, 2026.
- [12] J. C. Lagarias, The $3x + 1$ problem and its generalizations, *Amer. Math. Monthly* **92** (1985), no. 1, 3–23.
- [13] J. C. Lagarias (ed.), *The Ultimate Challenge: The $3x + 1$ Problem*, Amer. Math. Soc., Providence, RI, 2010.
- [14] E. M. Matveev, An explicit lower bound for a homogeneous rational linear form in logarithms of algebraic numbers, II, *Izv. Ross. Akad. Nauk Ser. Mat.* **64** (2000), no. 6, 125–180.
- [15] T. Oliveira e Silva, S. Herzog, and S. Pardi, Empirical verification of the $3x + 1$ and related conjectures, *The Ultimate Challenge: The $3x + 1$ Problem* (J. C. Lagarias, ed.), Amer. Math. Soc., 2010, pp. 189–207.
- [16] G. Rhin, Approximants de Padé et mesures effectives d’irrationalité, *Séminaire de Théorie des Nombres, Paris 1985–86*, Progress in Math., vol. 71, Birkhäuser, Boston, 1987, pp. 155–164.
- [17] D. J. Rudolph, $\times 2$ and $\times 3$ invariant measures and entropy, *Ergodic Theory Dynam. Systems* **10** (1990), 395–406.
- [18] J. Simons and B. M. M. de Weger, Theoretical and computational bounds for m -cycles of the $3n + 1$ problem, *Acta Arith.* **117** (2005), no. 1, 51–70.
- [19] R. P. Steiner, A theorem on the Syracuse problem, *Congr. Numer.* **20** (1977), 553–559.
- [20] T. Tao, Almost all orbits of the Collatz map attain almost bounded values, *Forum Math. Pi* **10** (2022), Paper No. e12, 56 pp.
- [21] R. Terras, A stopping time problem on the positive integers, *Acta Arith.* **30** (1976), no. 3, 241–252.
- [22] H. Weyl, Über die Gleichverteilung von Zahlen mod. Eins, *Math. Ann.* **77** (1916), no. 3, 313–352.
- [23] G. J. Wirsching, *The Dynamical System Generated by the $3n + 1$ Function*, Lecture Notes in Math., vol. 1681, Springer-Verlag, Berlin, 1998.